

Computer Network (CT 702)

BCT IV/I

Binod Sapkota

Asst. Professor

Department of Electronics and Computer Engineering

TU, IOE, Thapathali Campus, Kathmandu

1. Introduction to Computer Network

1.1 Computer network, importance and applications

1.2 Types of computer networks

1.3 Network topologies

1.4 Client/server and P2P networks

1.5 Protocols and standards, need of standardization

1.6 Layered network architecture, OSI and TCP/IP reference models

1.7 Addressing at different layers

1.8 Networking devices: Repeater, hub, bridge, switch, router, gateway

Networks

- The Internet is a giant network of networks.
- A network is a means to connect two or more computers together for the purposes of sharing information.
- A network may include PCs, and other devices like servers or printers.
- A network is connected through a communication channel.
- Early research was performed by the US Department of Defense in 1962.
 - This research group established ARPAnet (Advanced Research Project Agency) in order to connect the US Defense Department network.

What is a Computer Network?

- An interconnection of various computers to share software, hardware, resources and data through a communication medium between them.
- Any Computer Networking communication needs a sender, a receiver and a communication medium, protocols and operating system to establish networking and to transfer signal or Data from sender to the receiver.
- A networks model describes the organization of various computers in a network for using resources.
- Networks provide the benefits of *exchanging information or Data, sharing resources, reducing system costs, increased reliability and flexible working environment.*
- Every network involves hardware and software that connects computers and tools.

1.1 Uses of Computer Networks

Fundamental benefits/pros of using Computer Networking:

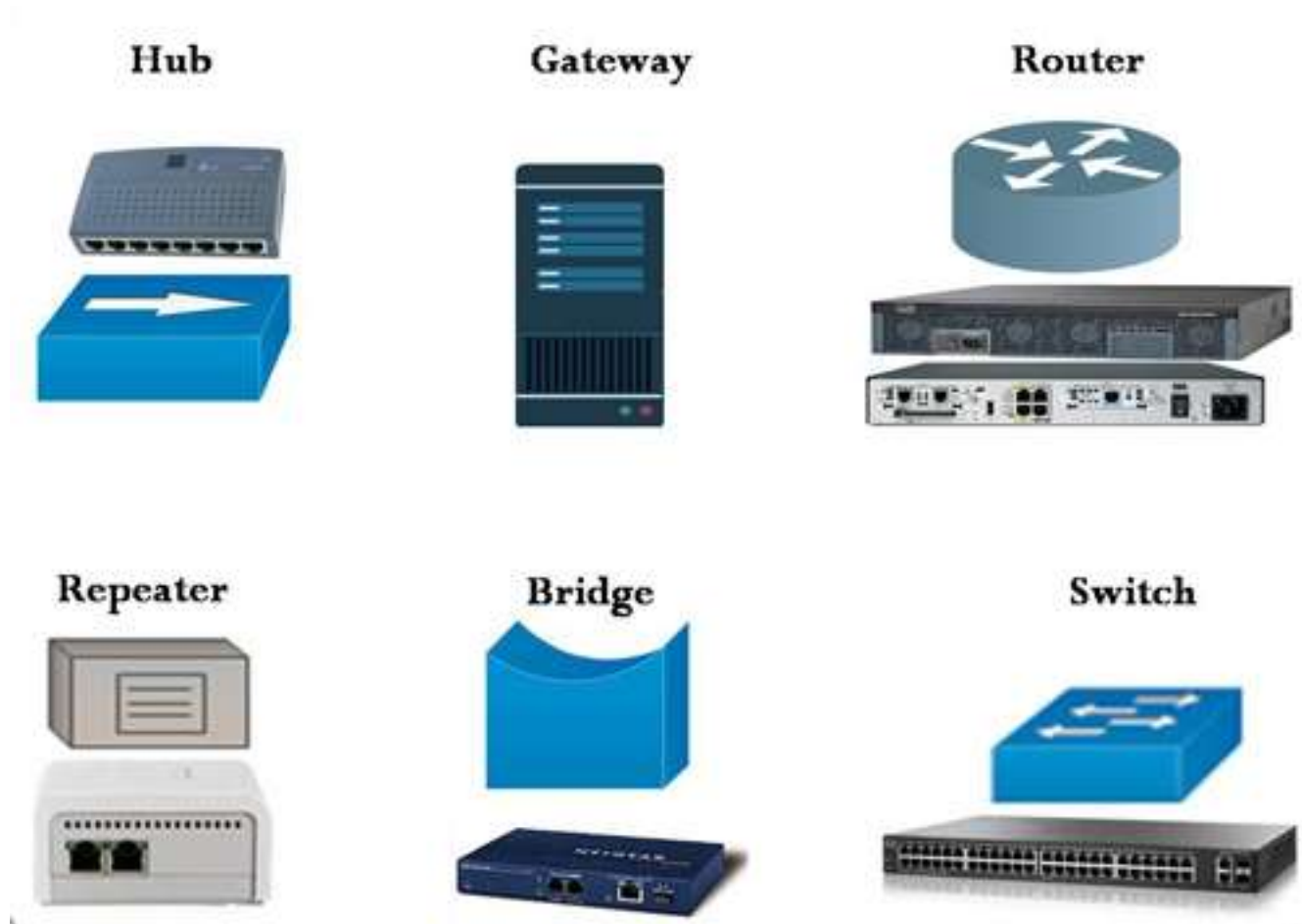
- to connect with multiple computers together to send and receive information when accessing the network.
- to share expensive software's and database among network participants.
- Provides fast and effective communication from one computer to another computer.
- to exchange/share data and information among users via a network at very fast speed.
- to share printers, scanners, and email.
- Electronic communication is more efficient and less expensive than without the network.

Application/**Need for Computer Networking**

- **File sharing**: to share data files.
- **Hardware sharing**: printers, scanners, CD-ROM drives, hard drives etc. Without computer networks.
- **Application sharing**: to implement client/server applications
- **User communication**: to communicate using e-mail, newsgroups, and video conferencing etc.
- **Network gaming**: multi-users to play games from different locations.
- **Voice over IP (VoIP)**: to send telephone calls (voice data) using standard Internet Protocol (IP) rather than by traditional PSTN.

Computer Network Components_(ch-4)

- Repeater
- Hub
- Bridge
- Switch
- Router
- Gateway



Network Architectures

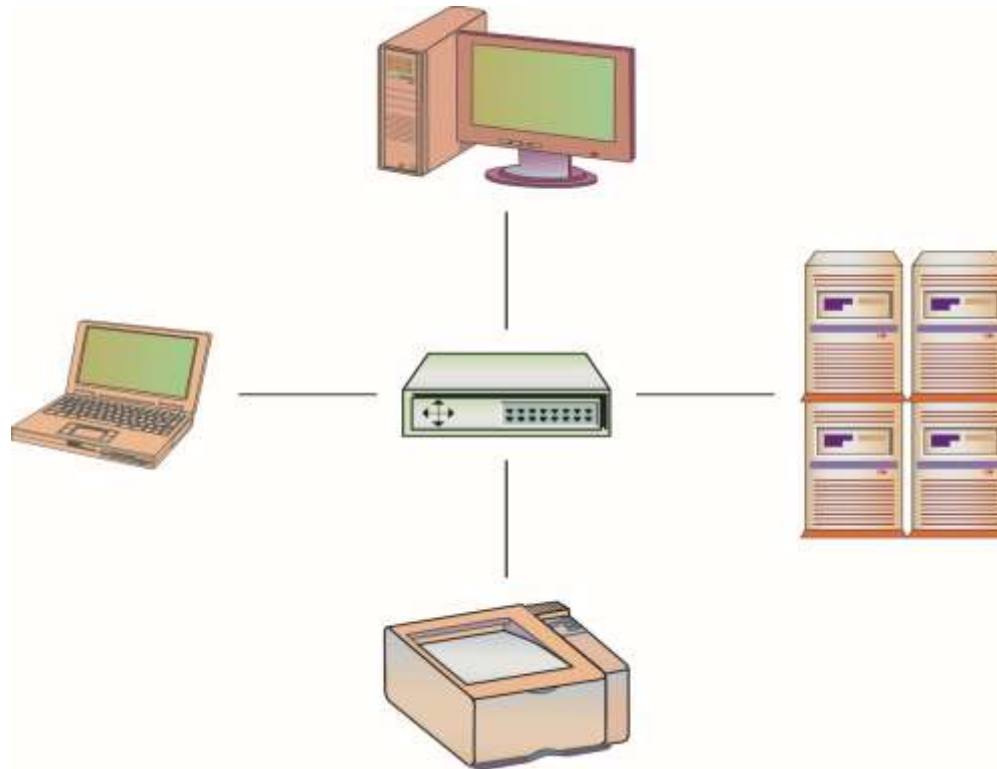
- network's architecture involves identifying the network physical configuration, logical operation, structure, procedures, data formats, protocols, and other components.
- local area network (LAN) is typically smaller in terms of size and geographic coverage, and consists of two or more connected devices. Home users
- wide area network (WAN) tends to be larger, covering more geographic area, and consists of two or more systems in geographically separated areas connected by any of a variety of methods.
 - Such as leased lines, radio waves, satellite relays, microwaves, or even dialup connections.

Network Topology

- One major component of every network's architecture is the network's topology—how the network is physically or logically arranged.
- The main classes of network topologies are star, ring, bus, and mixed.

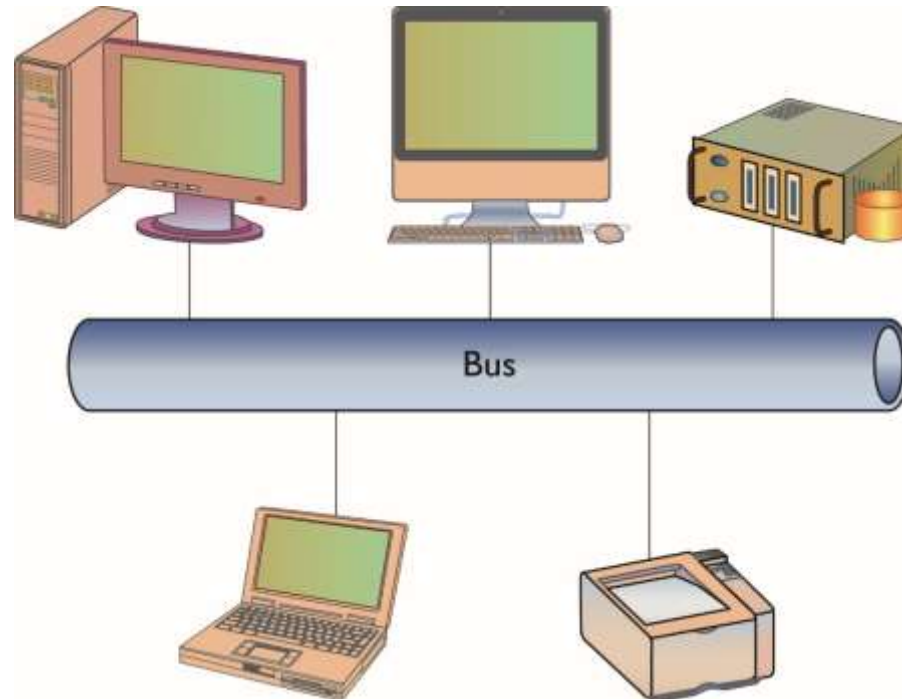
Star Topology

- Network components are connected to a central point.



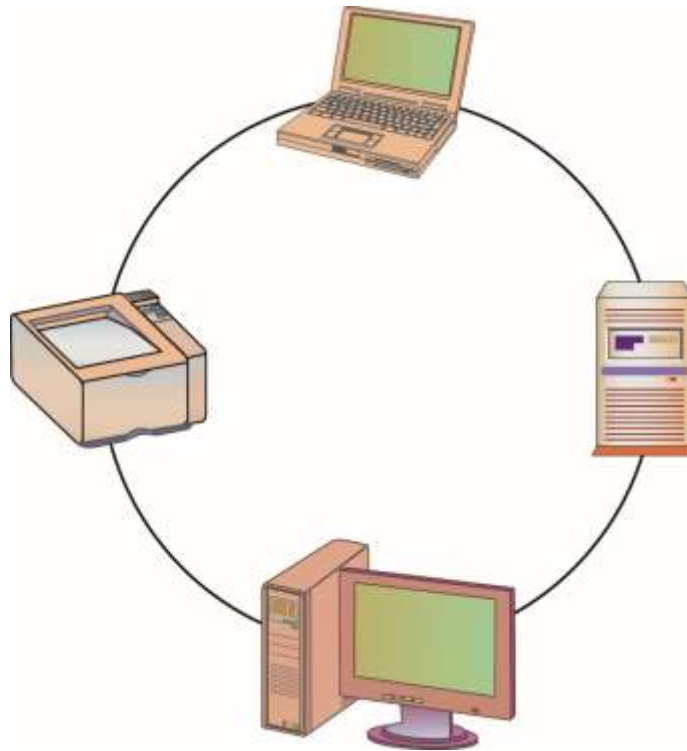
Bus Topology

- Network components are connected to the same cable, often called “the bus” or “the backbone.”



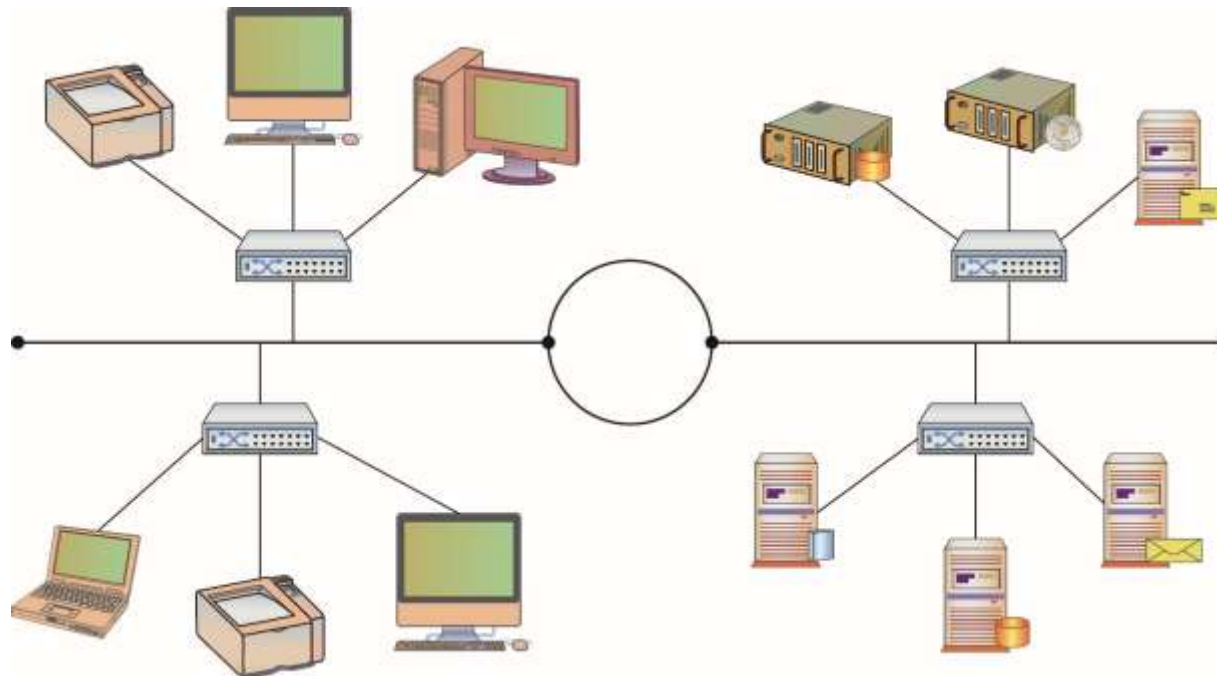
Ring Topology

- Network components are connected to each other in a closed loop with each device directly connected to two other devices.



Mixed Topology

- Larger networks, such as those inside an office complex, may use more than one topology at the same time. This is called a mixed topology or hybrid topology.



1.2 Types of Computer Networks

Computer networks can be logically classified into

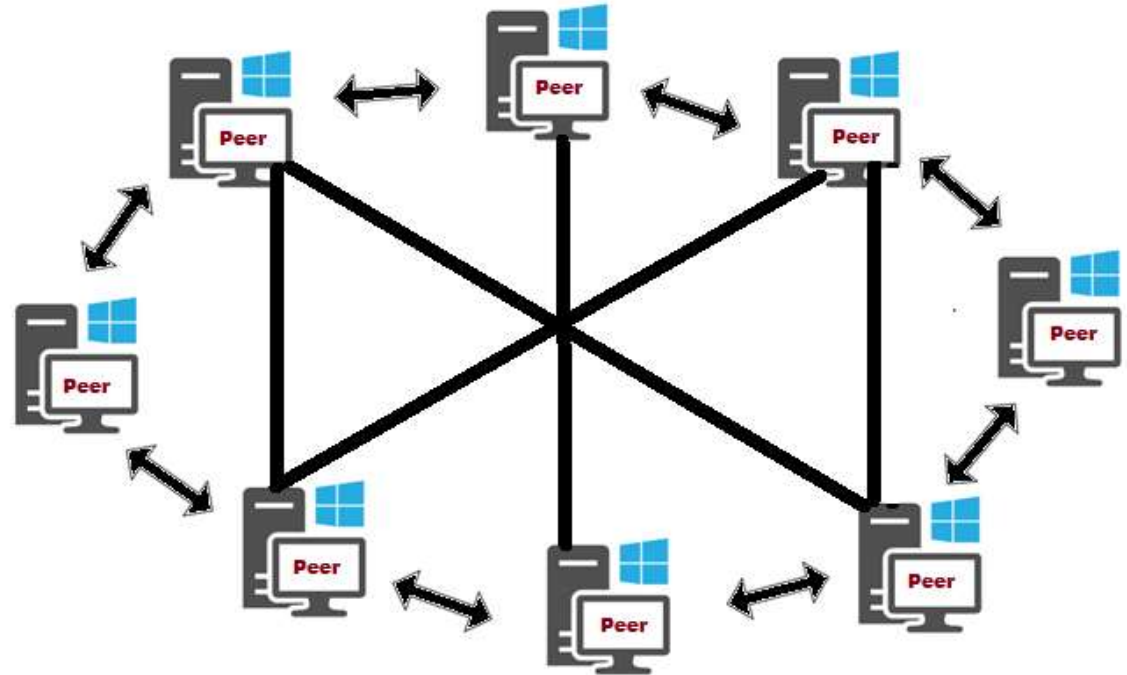
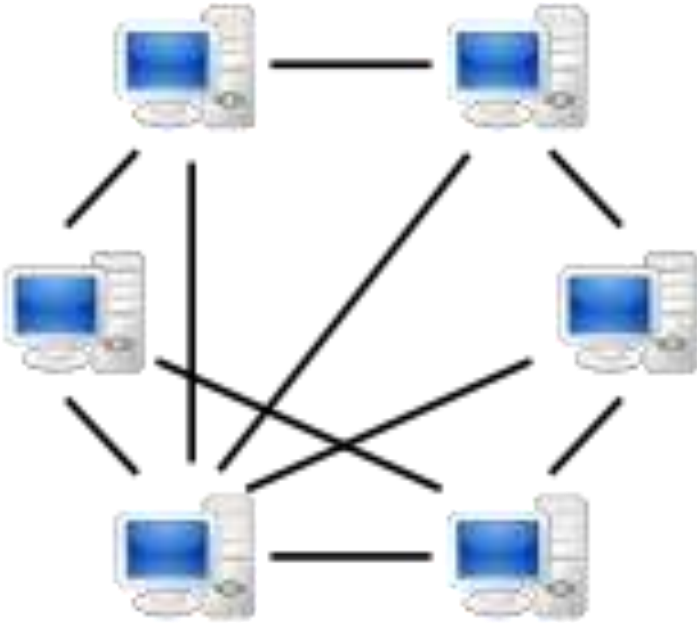
- 1) peer to peer networks (P2P) and
- 2) client server networks

- P2P network operating systems (P2PNOS) allow users to share resources and files located on their computers and to access shared resources found on other computers. However, they do not have a file server or a centralized management source .
- all computers are considered equal; they all have the same abilities to use the resources available on the network.
- designed primarily for small to medium local area networks. Nearly all modern desktop operating systems, such as Macintosh OSX, Linux, and Windows, can function as peer-to-peer network operating systems.

Peer to peer networks

- has no dedicated servers.
- number of workstations (or clients) are connected together for sharing devices, information or data. All the workstations (clients) are considered equal.
- Any one computer can act as client or server at any instance.
- This network is ideal for small networks where there is no need for dedicated servers, like home networks, small business networks, or retail shops.
- The Microsoft term for peer to peer network is “Workgroup”.
- There is no limitation for the number of computers in a peer to peer network.
- But peer to peer implementations are meant for small networks. Typically, a Workgroup contain less than 10 workstations.

Peer to peer networks



A peer-to-peer (P2P) network in which interconnected nodes ("peers") share resources amongst each other without the use of a central server

Peer to peer networks

- The main **disadvantages** of peer to peer networks are listed below.
 - Everything is kept distributed in different computers.
 - User generated files are stored in individual computers. Data backup is extremely difficult.
 - Each computer has its own user database.
 - There is no centralized user & user privilege management. Users need to remember their user ids and passwords in every computer.
 - Managing network users is extremely difficult.

Client server networks

- A client server computer network model is made-up of client computers and server computers.
- **What is a client computer?**
 - A computer which is seeking any resource from another computer is a client computer.
 - For Example: Downloading a file from a File Server, Browsing Intranet/Internet etc.
 - The network user normally uses a client computer to perform his day to day work.
- **What is a server computer?**
 - If a computer has a resource which is served to another computer, it is a server computer.
 - The client establishes a connection to a server and accesses the services installed on the server.
 - A server computer is installed with appropriate Operating System and related software to serve the network clients with one or more services.

Client server networks

- In a client server network, high-end servers, installed with the **Network Operating System (Server Operating System)** and the related software, serve the clients continuously on a network, by providing them with specific services upon request.
- Client/server network operating systems allow the network to centralize functions and applications in one or more dedicated file servers.
- The file servers become the heart of the system, providing access to resources and providing security. Individual workstations (clients) have access to the resources available on the file servers.
- The network operating system provides the mechanism to integrate all the components of the network and allow multiple users to simultaneously share the same resources irrespective of physical location.
- **UNIX/Linux** and the **Microsoft family of Windows Servers** are examples of client/server network operating systems.
- A server computer is built and installed to run continuously 24/7 without a break. An unexpected shutdown of a server computer can cause catastrophic damage to a business.

Advantages/disadvantages of a client/server network

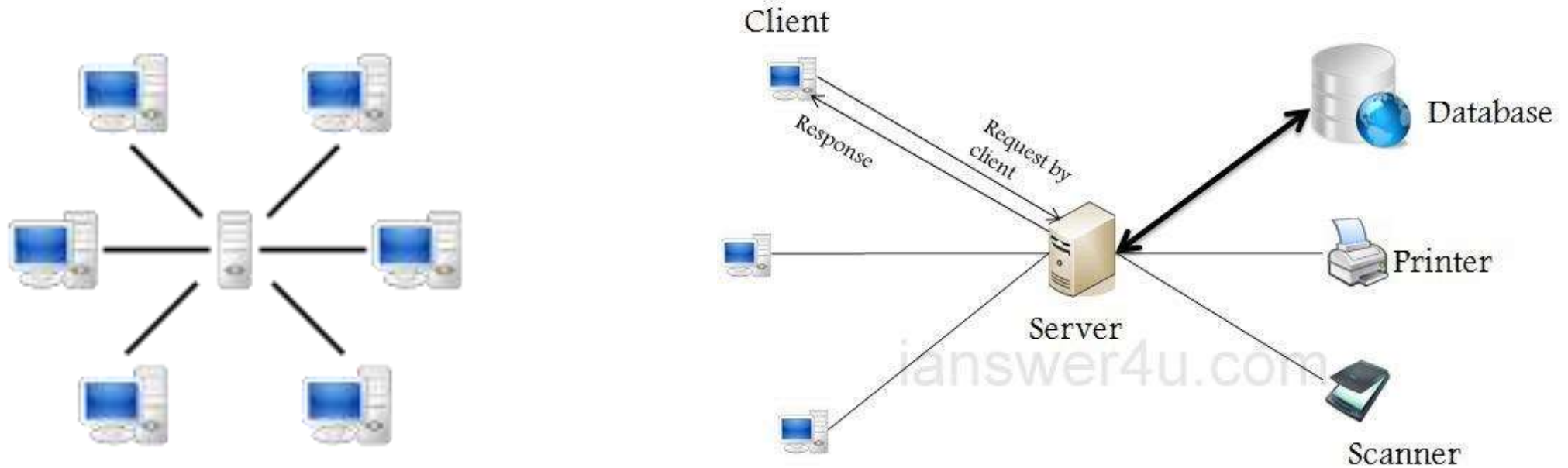
- **Advantages:**

- Centralized - Resources and data security are controlled through the server.
- Scalability - Any or all elements can be replaced individually as needs increase.
- Flexibility - New technology can be easily integrated into system.
- Interoperability - All components (client/network/server) work together.
- Accessibility - Server can be accessed remotely and across multiple platforms.

- **Disadvantages:**

- Expense - Requires initial investment in dedicated server.
- Maintenance - Large networks will require a staff to ensure efficient operation.
- Dependence - When server goes down, operations will cease across the network.

Client server networks



A network based on the [client-server model](#), where individual [clients](#) request services and resources from centralized [servers](#)ized administrative system

Client server networks

- Building a client server based network is costlier than building a peer to peer network.
- Client server based networks require dedicated servers.
- Server hardware is costlier than normal desktop client computer hardware.
- Network Operating Systems (Server Operating System) are also costlier than client Operating Systems (Desktop Operating Systems).

Client server networks

- Different types of servers used in client server-based networks are:
 - Database server
 - Application Server
 - DNS server
 - Web server
 - Messaging server
 - File Server
 - Print server
 - Mail server

Network Operating Systems - NOS (Server Operating Systems)

- Operating System controls almost all the resources in a computer, including networks, data storage, user & user password database, peripheral devices etc.
- Most popular Network Operating Systems are listed below.
 - Microsoft Windows Server 2012
 - Microsoft Windows Server 2016
 - Microsoft Windows Server 2019 (Current)
 - Unix (Oracle Solaris, IBM AIX, HP UX, FreeBSD, NetBSD, OpenBSD, XinuOS Open Server/SCO Unix etc.)
 - GNU/Linux (RedHat Enterprise Linux, Debian Linux, SUSE Linux Enterprise Server, Ubuntu Server, CentOS Server, Fedora Server etc.)

Client Operating Systems (Workstation Operating Systems, or Desktop Operating Systems)

- Most popular Client Workstation Operating Systems are listed below.
 - Windows 95/98/ME Vista (obsolete)
 - Windows NT Workstation / Windows 2000 Professional (obsolete)
 - Windows XP (obsolete)
 - Windows 7 (obsolete)
 - Windows 8 / Windows 8.1 (obsolete)
 - Windows 10 Desktop (Current)
 - Red Hat Enterprise Linux Workstation
 - SUSE Linux Enterprise Desktop
 - Ubuntu Desktop
 - LinuxMint
 - Fedora Workstation
 - elementary OS

1.3 Protocols and Standards

- **Network Protocol**: a common network communication standard, which is used to define a method of exchanging data over a computer network.
- Network protocols define **rules and conventions** for communication between different devices, participating in a computer network.
- A network protocol is that set of rules that should be followed by different computers or computer-like devices when they participate in network communication.
- A protocol is “**A precise set of standards that governs communications between computers on a network.**”
- **Protocol suites** are collections of protocols that enable network communication between hosts.
- Protocols determine the format, timing, sequencing, and error control in data communication.
- Without protocols, the computer cannot make or rebuild the stream of incoming bits from another computer into the original format

TCP/IP

- Two key protocols
 - Transmission Control Protocol (TCP)
 - Offers reliable delivery of messages of arbitrary size
 - Defines a robust delivery mechanism for all kinds of data across a network.
 - Internet Protocol (IP)
 - Manages the routing of network transmissions

Standards Groups That Oversee TCP/IP

- Internet Society (ISOC)
- Internet Architecture Board (IAB)
- Internet Engineering Task Force (IETF)
 - Responsible for creating and managing RFCs, in which the rules and formats for all related protocols and services are described.
- Internet Research Task Force (IRTF)
- Internet Corporation for Assigned Names and Numbers (ICANN)

TCP/IP Standards and RFCs

- Almost all protocols used in the Internet are common standards defined by RFC (Request for Comments).
- Request for Comments (RFCs)
 - Provide documentation to understand, implement, and use TCP/IP protocols
- Index for all RFCs available at
 - www.faqs.org/rfcs/
- RFC 2026
 - Describes how an RFC is created
- Examples of standard network protocols are TCP (Transmission Control Protocol), UDP (User Datagram Protocol), IP (Internet Protocol), ARP (Address Resolution Protocol), HTTP (Hypertext Transfer Protocol), FTP (File Transfer Protocol), TFTP (Trivial File Transfer Protocol), SMTP (Simple Mail Transfer Protocol), SSH (Secure Shell), Telnet etc.

1.4 OSI model and TCP/IP model

- OSI reference model
 - A network reference model
 - Formally known as ISO/OSI –International organization for standardization Open systems Interconnection
 - Designed in 1980 to replace TCP/IP
 - Standard way to explain how networks operate
 - TCP/IP is the open standard protocol suite of choice

Breaking Networking into Layers

- Divide-and-conquer approach
 - Separates networking hardware concerns from those related to networking software
- Key points about **networking**
 - Easier to solve problems when broken into series of smaller problems
 - Layers operate independently of one another
 - Changes to one layer need not affect other layers because individual layer **encapsulate** specific

Breaking Networking into Layers (cont'd.)

- Key points about networking
 - Individual layers work together on pairs of computers
 - Different expertise is needed at each layer
 - Layers in a network implementation work together to create a general solution
 - Network protocols usually map into one or more layers
 - TCP/IP itself is designed around a layered model for networking

How Protocol Layers Behave

- Layers
 - Exist to encapsulate or isolate specific types of functionality
 - Provide services to the layer above
 - Deliver data to or accept data from the layer below
- Protocol data units (PDUs)
 - Include “envelope information” in the form of specific headers and trailers

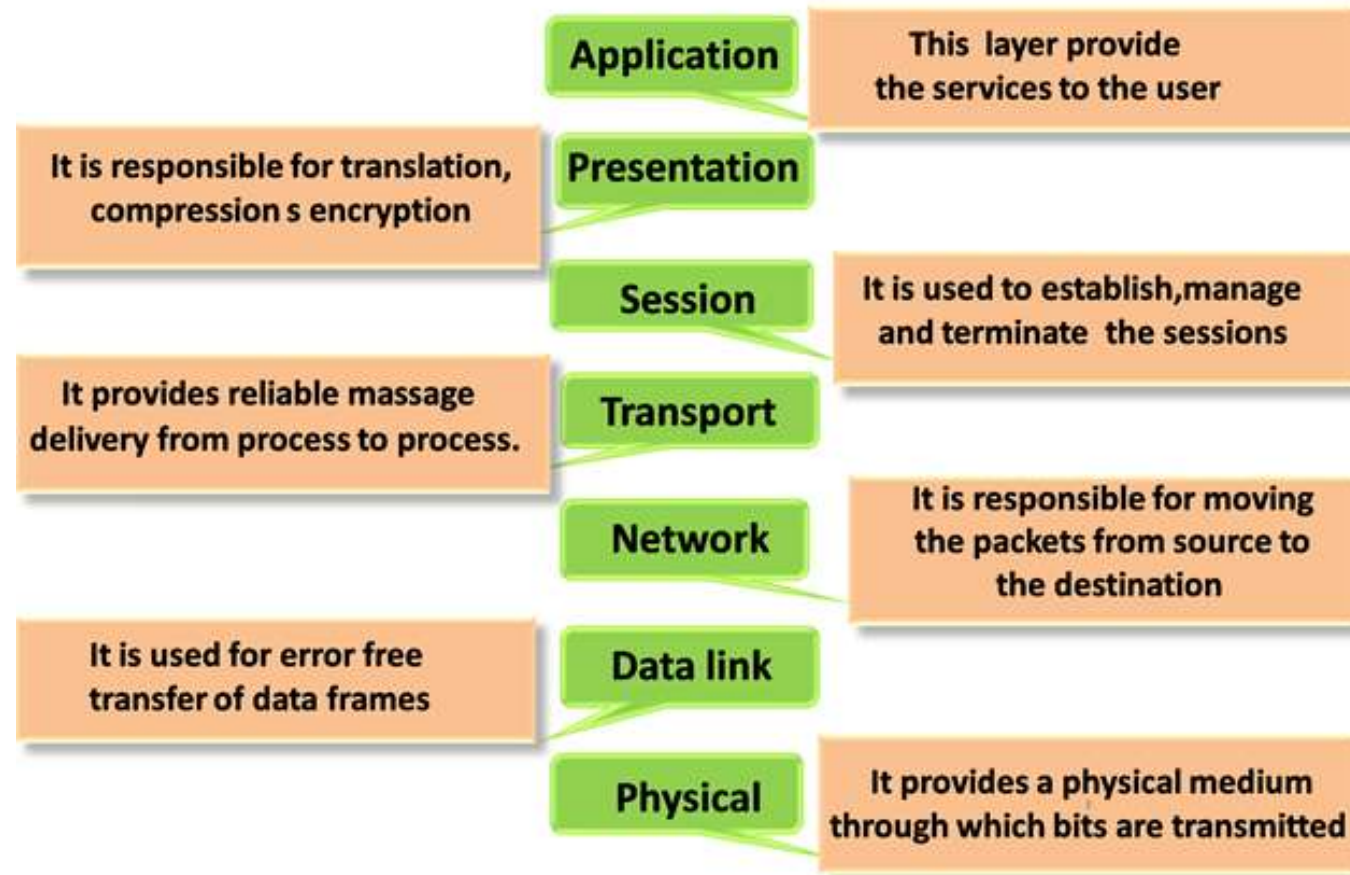
OSI Reference Model

- Is a framework that is used to understand how information travels throughout a network.
- Explains how packets travel through the various layers to another device on a network, even if the sender and destination have different types of network media.
- * All People Seem To Need data processing

OSI Reference Model

- In the OSI reference model, there are seven numbered layers, each of which illustrates a particular network function.
- **Advantages:**
- It breaks network communication into smaller, more manageable parts.
- It standardizes network components to allow multiple vendor development and support.
- It allows different types of network hardware and software to communicate with each other.
- It prevents changes in one layer from affecting other layers.
- It divides network communication into smaller parts to make learning it easier to understand.

OSI layer



OSI Reference Model

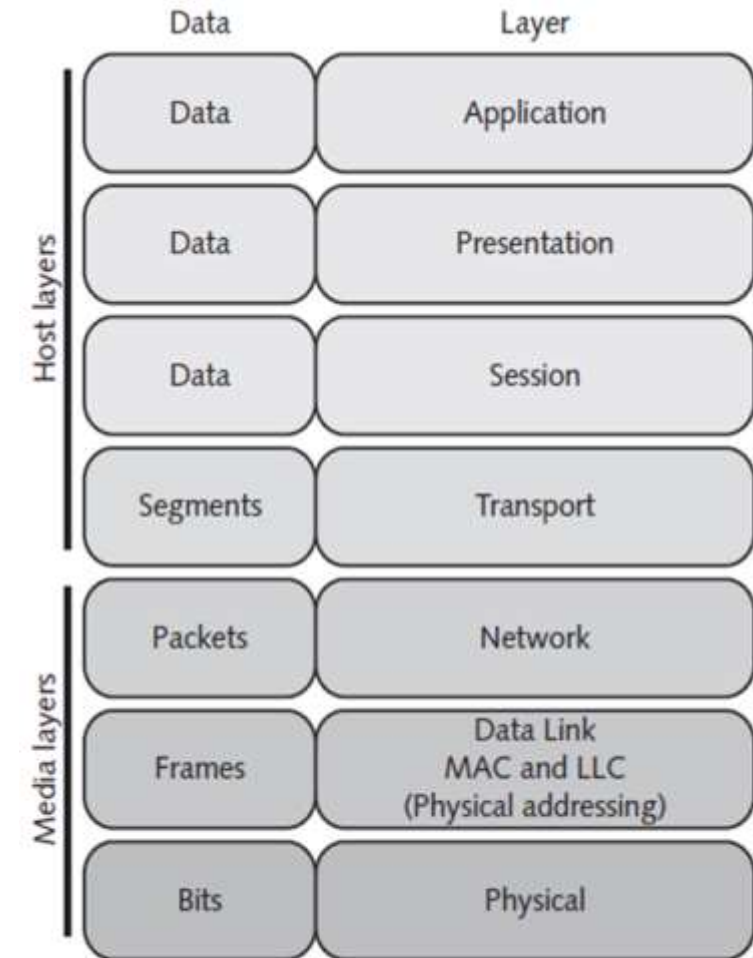
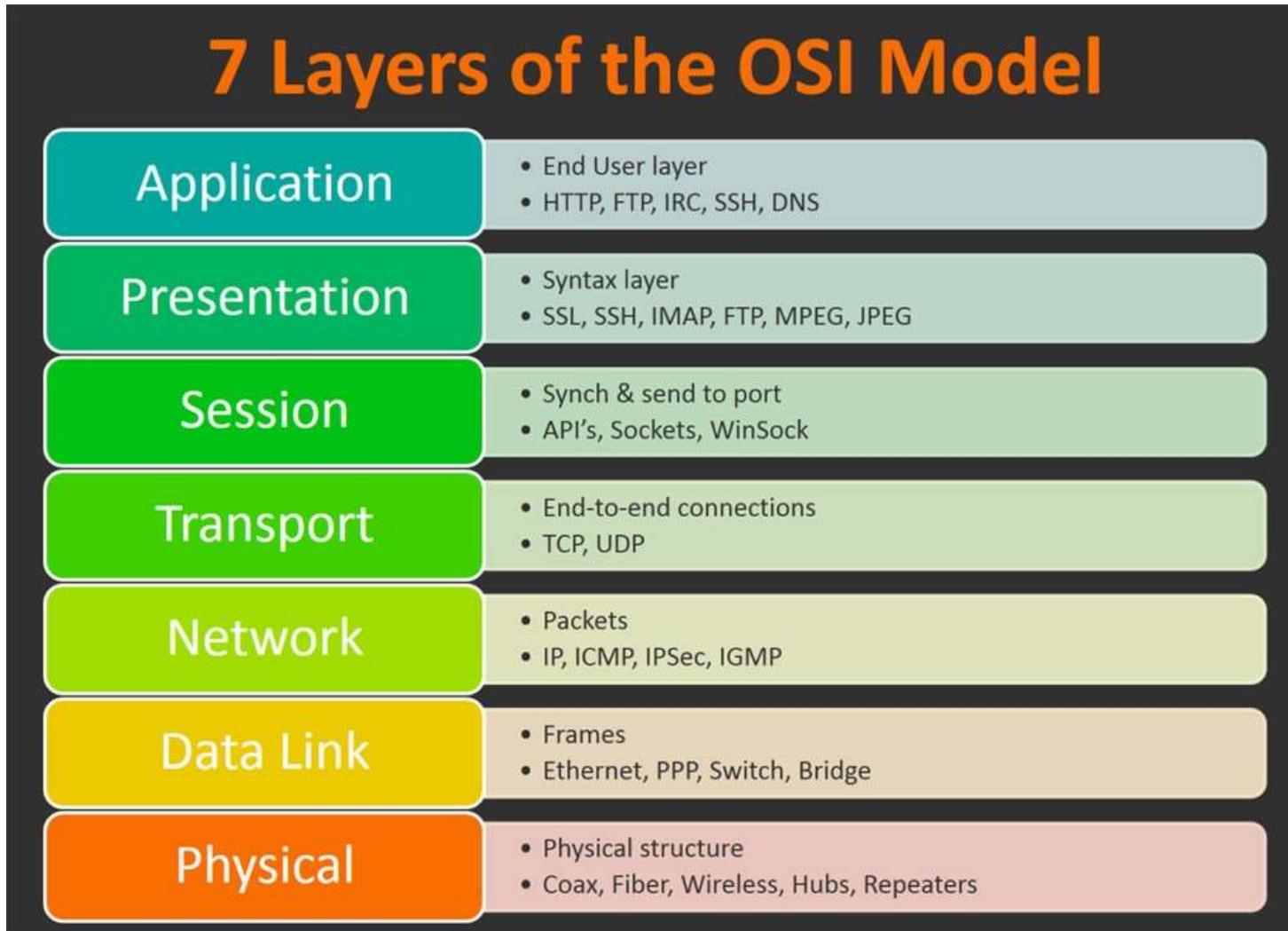
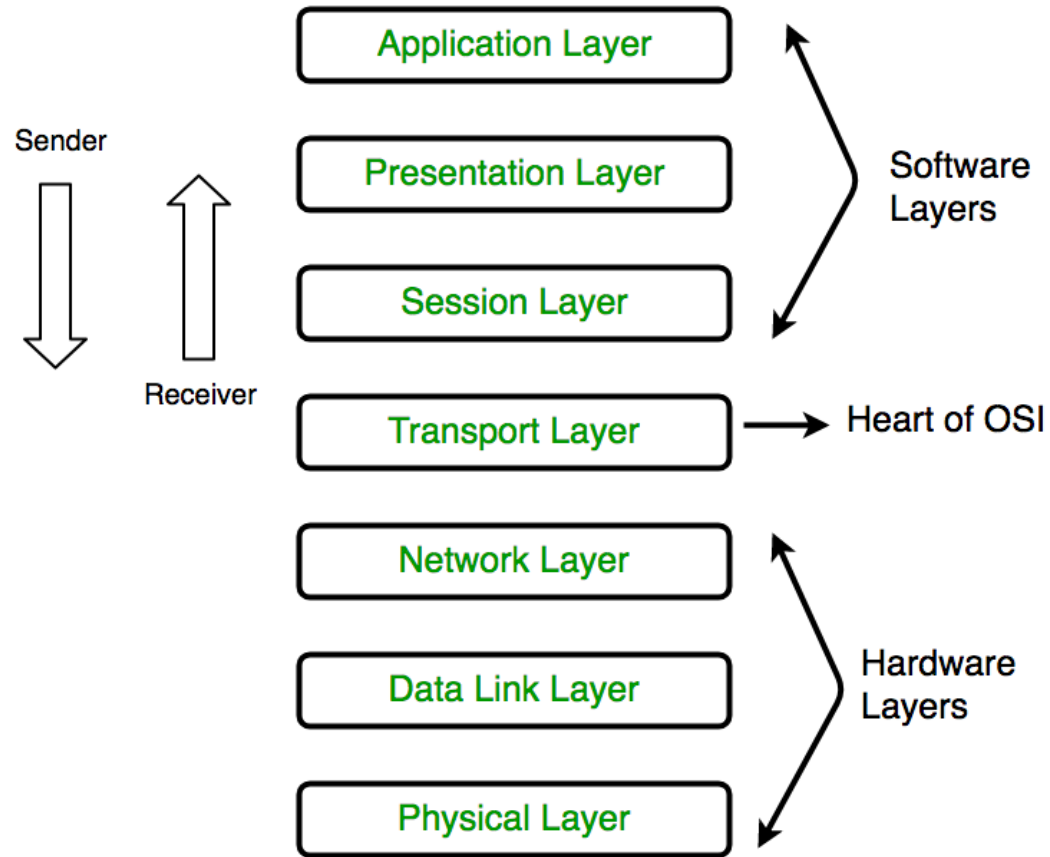


Figure 1-1 ISO/OSI network reference model layers

Source: Figure adapted from illustration at ccnaGuru.com

7 OSI layers

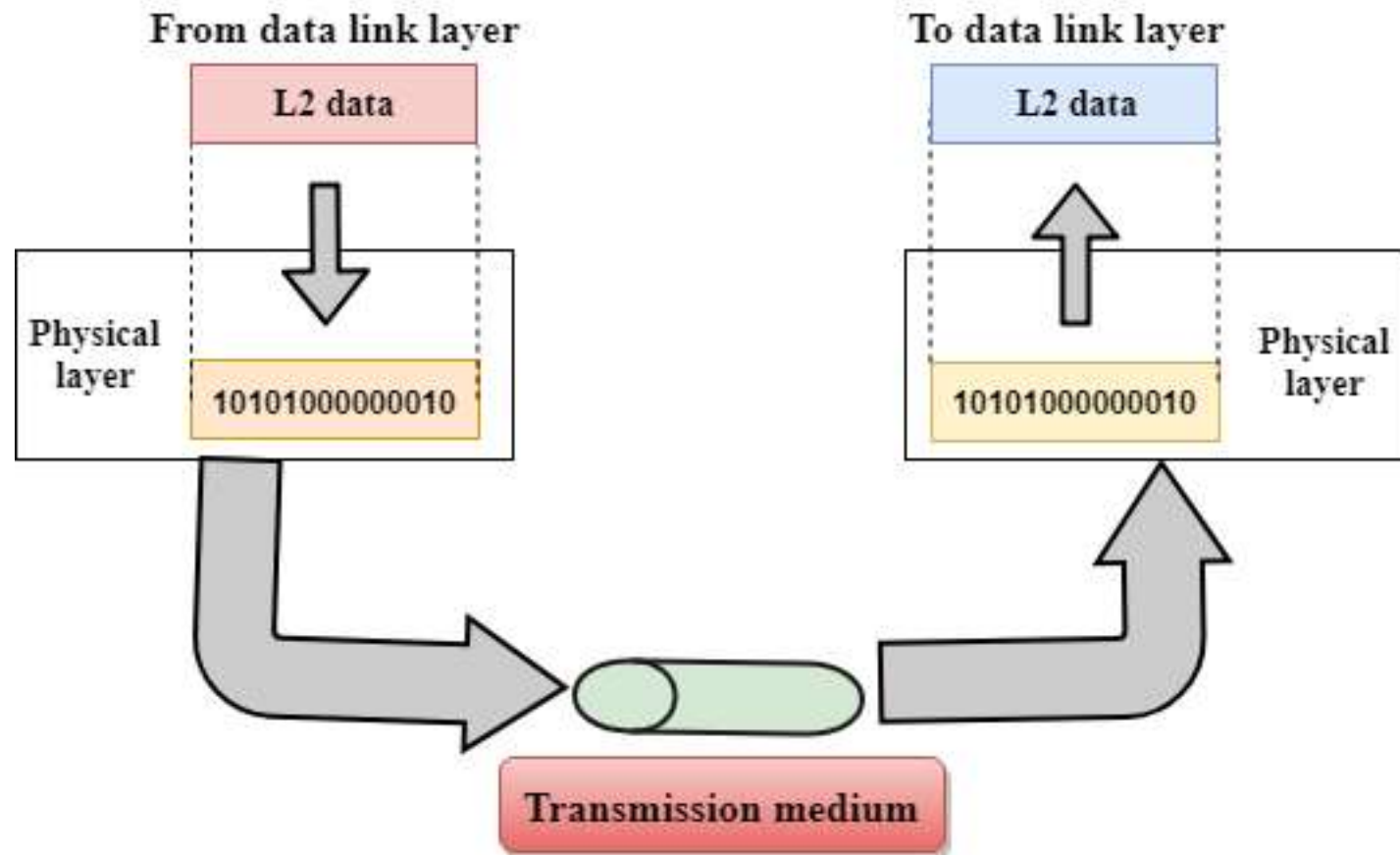


OSI model		
Layer	Name	Example protocols
7	Application Layer	HTTP, FTP, DNS, SNMP, Telnet
6	Presentation Layer	SSL, TLS
5	Session Layer	NetBIOS, PPTP
4	Transport Layer	TCP, UDP
3	Network Layer	IP, ARP, ICMP, IPSec
2	Data Link Layer	PPP, ATM, Ethernet
1	Physical Layer	Ethernet, USB, Bluetooth, IEEE802.11

Physical layer

- Physical layer of the OSI model is only concerned with
 - the physical characteristics of **electrical** or **optical signaling** techniques which includes the voltage of the electrical current used to transport the signal,
 - the media type (**Twisted Pair, Coaxial Cable, Optical Fiber** etc.),
 - transmitting raw bits over communication channel, data transmission or data rate
 - impedance characteristics,
 - physical shape of the connector,
 - Synchronization etc.
 - Job is to activate, maintain, and deactivate network connections
- The Physical Layer is limited to the processes needed to place the communication signals over the media, and to receive signals coming from that media.
- The lower boundary of the physical layer of the Open Systems Interconnection (OSI) model is the physical connector attached to the transmission media
- Protocol Data Unit(PDU): **bps**

Physical layer



Functions of a Physical layer

- **Line Configuration:** It defines the way how two or more devices can be connected physically.
- **Data Transmission:** It defines the transmission mode whether it is simplex, half-duplex or full-duplex mode between the two devices on the network.
- **Topology:** It defines the way how network devices are arranged.
- **Signals:** It determines the type of the signal used for transmitting the information.
- It establishes, maintains and deactivates the physical connection.
- It specifies the mechanical, electrical and procedural network interface specifications.

Data Link layer

- **Job** is to
 - Enable **reliable transmission** of data through the Physical layer at the sending end
 - Check reliability at the receiving end
- There is a direct **logical link control** on the network
 - Creating temporary point to point links between a pair of MAC layer addresses to enable data transfer
- Manages **point-to-point** transmission across the networking medium
 - Shipping data from one MAC layer address to another
- Protocol Data Unit(PDU): **Frame**

Datalink layer

- Datalink layer is responsible for providing end-to-end validity of the data being transmitted.
- This layer is responsible for the error-free transfer of data frames.
- It defines the format of the data on the network.
- It provides a reliable and efficient communication between two or more devices.
- It is mainly responsible for the unique identification of each device that resides on a local network.
- The Data Link Layer is logically divided into two sublayers,
 - The Media Access Control (MAC) Sublayer and
 - the Logical Link Control (LLC) Sublayer.

Datalink layer

- The **MAC sub-layer** determines/maintains **MAC address/physical address/hardware address of the hosts** for communicating with other devices on the network.
 - managing access to whatever medium is in use
- A MAC layer is a link between the Logical Link Control layer and the network's physical layer.
- It is used for transferring the packets over the network.
- **Logical Link Control Layer**
 - It is responsible for transferring the packets to the Network layer of the receiver that is receiving.
 - It identifies the address of the network layer protocol from the header.
 - The **Logical Link Control** sublayer is responsible for **synchronizing frames, error checking, and flow control.**

Functions of the Data-link layer

- **Framing**
- **Physical Addressing**
- **Error detection/Control**
- **Flow Control:** Flow control is the main functionality of the Data-link layer. It is the technique through which the constant data rate is maintained on both the sides so that no data get corrupted. It ensures that the transmitting station such as a server with higher processing speed does not exceed the receiving station, with lower processing speed.
- **Access Control:** When two or more devices are connected to the same communication channel, then the data link layer protocols are used to determine which device has control over the link at a given time.

Data Link layer services

- WAN encapsulation of **frames** at Data Link layer involves one or more of the following services
 - Addressing, a unique destination is required
 - Bit-level integrity check
 - Delimitation, information boundary for data link frame
 - Protocol identification (PID)
- *Layer 2 Data Link examples include PPP, FDDI, ATM, IEEE 802.5/ 802.2, IEEE 802.3/802.4, HDLC, Frame Relay.*

Frame Types and Sizes _{ch-3}

- At Data Link layer
 - Protocol data units (PDUs) are called **frames**
- Frame
 - Represents same data that appears in digital form at the Network layer in an IP datagram in the form of whatever sequence of electrical signals map to that data.
- Ethernet frame types that TCP/IP can use
 - **Ethernet II**
 - **Ethernet 802.2 Logical Link Control**

Network layer

- Handles **logical addresses** associated with individual machines on a network
- Uses **addressing** information to
 - Determine how to send a **PDU**
- Embodies notion of multiple simultaneous connections between different IP addresses
- Flexible enough to
 - Recognize and use multiple routes between a sender and a receiver
- It determines the best path to move data from source to the destination based on the network conditions, the priority of service, and other factors.
- Responsible for routing and forwarding the packets.
- Routers are the layer 3 devices, they are specified in this layer and used to provide the routing services within an internetwork.
- PDU: **Packet**

Network layer

- The Network layer of the OSI model is responsible for managing **logical addressing** information in the packets and the delivery of those packets to the correct destination.
 - The **logical address** are used to uniquely identify a computer on the network, but at the same time identify the network that system resides on.
 - The **logical address** is used by network layer protocols to deliver the packets to the correct network.
 - The **logical addressing system** used in Network Layer is known as **IP address**.
- **Routers**, which are special devices used to build the network, direct the data packet generated by Network Layer using information stored in a table known as routing table.
 - The routing table is a list of available destinations that are stored in memory on the routers.

Functions of Network Layer

- **Internetworking**: An internetworking is the main responsibility of the network layer. It provides a logical connection between different devices.
- **Addressing** : A Network layer adds the source and destination address to the header of the frame. Addressing is used to identify the device on the internet.
- **Routing**: Routing is the major component of the network layer, and it determines the best optimal path out of the multiple paths from source to the destination.
 - Controls the operation of the subnet, routing
- **Packetizing**: A Network Layer receives the packets from the upper layer and converts them into packets. This process is known as Packetizing. It is achieved by internet protocol (IP).
- error handling, congestion control and packet sequencing.

Transport Layer

- Ensures reliable end-to-end transmission of PDUs
- Includes end-to-end error detection and error recovery
- Segmentation
 - Involves cutting up a big message into a numbered sequence of chunks called ***segments***
- PDUs used at the Transport layer are called ***segments***, or *data segments*

Functions of Transport Layer

- Provides:
- quality of service (QoS) functions and ensures the complete delivery of the data
- Segmentation and Reassembly.
- Connection Control
- Flow Control
- Error Control
- Session multiplexing
- The integrity of the data is guaranteed at this layer via error correction and similar functions.

TCP establishes and maintains a connection from the client to the web server, and packages the higher-layer data into segments. A sequence number is assigned to each segment so that data can be reassembled upon arrival.

Two Protocols

- **Transmission Control Protocol (TCP)**

- It is a standard protocol that allows the systems to communicate over the internet.
- It establishes and maintains a connection between hosts.
- When data is sent over the TCP connection, then the TCP protocol divides the data into smaller units known as segments.
- Each segment travels over the internet using multiple routes, and they arrive in different orders at the destination.
- The transmission control protocol reorders the packets in the correct order at the receiving end.

- **User Datagram Protocol (UDP)**

- It is an **unreliable** transport protocol as in this case receiver does not send any acknowledgment when the packet is received, the sender does not wait for any acknowledgment. Therefore, this makes a protocol unreliable.
- TCP (Transmission Control Protocol) and UDP (User Datagram Protocol) uses a mechanism known as "**Port Number**" to enable multiplexing and de-multiplexing.
- Port numbers identify the originating network application on the source computer and destination network application on the receiving computer.

Session Layer

- The session layer sets up, coordinates, and terminates conversations, exchanges, and dialogues between the applications at each end.
- It deals with session and connection coordination.
- Defines mechanisms to
 - Permit senders and receivers to request that a conversation start or stop
 - Keep a conversation going even when traffic may not otherwise flow between the parties involved
- Checkpoints
 - Define the last point up to which successful communications are known to have occurred
- *Session examples include NFS, Net Bios names, RPC, SQL.*

Session Layer

- Handles authentication and authorization functions.
- also manages the connection between the two communicating end points, establishing a connection, maintaining the connection, and ultimately terminating it.
- It provides: Session establishment, maintenance and termination:
 - allows two application processes on different machines to establish, use and terminate a connection, called a session.
 - Session support: performs the functions that allow these processes to communicate over the network, performing security, name recognition, logging, and so on.

Presentation layer

- Handles transforming data from Generic, **network-oriented forms** of expression to more specific, **platform-oriented forms** of expression
- When the presentation layer receives data from the application layer, to be sent over the network, or receives network data from the session layer, it makes sure that the data is in the proper format.
 - If it is not, the presentation layer converts the data to the proper format.
- **Masks the differences in data formats between dissimilar systems.**
- **Encodes and decodes data, encrypts and decrypts data, and compresses and decompresses data.**

Presentation layer

- **formats the data** to be presented to the application layer.
- can be viewed as the **translator** for the network.
- The presentation layer provides:
 - **Character code translation**: for example, ASCII to EBCDIC.
 - **Data conversion**: bit order, **CR-CR/LF**, integer-floating point, and so on.
 - **Data compression**: reduces the number of bits that need to be transmitted on the network.
 - **Data encryption**: encrypt data for security purposes. For example, password encryption.
- *Presentation examples include encryption, ASCII, EBCDIC, TIFF, GIF, PICT, JPEG, MPEG, MIDI.*

Application Layer

- Defines an interface that applications can use to request **network services**/Application for the user for network use
- Real traffic data will be often generated from the Application Layer
- Defines a set of access controls over the network
- Serves as the window for users and application processes to access network services
- Email, *WWW browsers*, *NFS*, *SNMP*, *Telnet*, *SMTP*, *HTTP*, *FTP*

Application Layer

- Supports application and end-user processes.
- Communication partners are identified, quality of service is identified, user authentication and privacy are considered, and any constraints on data syntax are identified.
- Everything at this layer is application-specific. This layer provides application services for file transfers, e-mail, and other network software services.
- *Application protocols: DHCP,DNS,FTP,HTTP,IMAP & IMAP4, POP& POP3,SMTP,Telnet,SNMP*

Application Layer

- Contains a variety of commonly needed functions:
 - Resource sharing and device redirection
 - Remote **File transfer, access, and management (FTAM)**, printer access
 - Inter-process communication
 - Network management
 - Directory services
 - Electronic messaging (such as mail)
 - Network virtual terminals
- *** The Internet can provide data in a wide variety of formats, a function of the Presentation layer. Common formats on the Internet include HTML, XML, PHP, GIF, and JPEG. Any encryption or compression mechanisms used on a website are also considered a Presentation layer function.

OSI model Protocol Suits

1. Physical layer

- EIA/TIA-232, EIA/TIA-449, ITU-T V-Series, I.430, I.431, PDH, SONET/SDH, PON, OTN, DSL, IEEE 802.3, IEEE 802.11, IEEE 802.15, IEEE 802.16, IEEE 1394, ITU-T G.hnPHY, USB, Bluetooth, RS-232, RS-449, 10BaseT, 10Base100, CSU/DSU, DCE and DTE

2. Data link layer

- ATM, ARP, IS-IS, SDLC, HDLC, CSLIP, SLIP, GFP, PLIP, IEEE802.2, LLC, MAC, L2TP, IEEE 802.3, Frame Relay, ITU-TG.hn DLL, PPP, X.25 LAPB, Q.921 LAPD, Q.922 LAPF, PPP

3. Network layer

- IP, IPv4, IPv6, ICMP, IPsec, IGMP, IPX, AppleTalk, X.25 PLP

4. Transport layer

- TCP, UDP, SCTP, DCCP, SPX

5. Session layer

- RPCs and NFS

6. Presentation layer

- ASCII, BMP, GIF, JPEG, WAV, AVI, and MPEG

7. Application layer

- NNTP, SIP, SSI, DNS, FTP, Gopher, HTTP, NFS, NTP, SMTP, SNMP, Telnet, DHCP, Netconf, POP

TCP/IP Networking Model

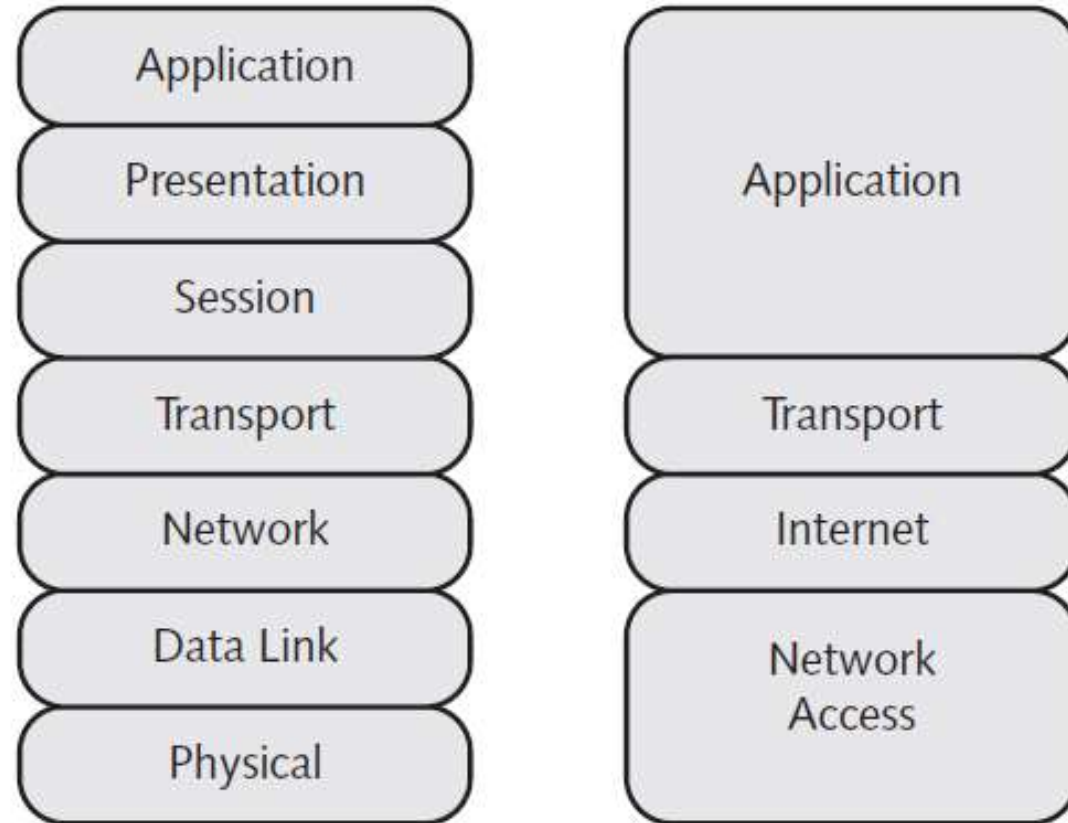


Figure 1-2 OSI reference model and TCP/IP networking model layers

What is TCP/IP

- Large collection of networking protocols and services
- A **protocol** is “A precise set of standards that governs communications between computers on a network.”
- **Protocol suites** are collections of protocols that enable network communication between hosts.
- The **Internet protocol suite** provides end-to-end data communication specifying how data should be packetized, addressed, transmitted, routed, and received.
- Protocols determine the format, timing, sequencing, and error control in data communication.
- Without protocols, the computer cannot make or rebuild the stream of incoming bits from another computer into the original format

TCP/IP Networking Model

- Design model that describes TCP/IP differs somewhat from OSI reference model
- The TCP/IP model was developed prior to the OSI model.
- TCP/IP is a hierarchical protocol made up of interactive modules, and each of them provides specific functionality.
- Transport layers for both models map together quite well as does the
 - **Network layer** from the OSI reference model and the **Internet layer** from the TCP/IP model

TCP/IP Network Access Layer

- A network layer is the lowest layer of the TCP/IP model.
- A network layer is the combination of the **Physical layer and Data Link** layer defined in the OSI reference model.
- It defines how the data should be sent physically through the network.
 - including how bits are electrically or optically signaled by hardware devices that interface directly with a network medium, such as coaxial cable, optical fiber, or twisted pair copper wire.
- This layer is mainly responsible for the transmission of the data between two devices on the same network.
- The functions carried out by this layer are encapsulating the **IP datagram** into **frames** transmitted by the network and mapping of IP addresses into physical addresses.
- The protocols used by this layer are ethernet, token ring, FDDI, X.25, frame relay and wireless media devices etc.

TCP/IP Network Access Layer Protocols

- PPP
 - Most important TCP/IP Network Access layer protocol
- PPPoE (“PPP over Ethernet”)
 - Widely used on Ethernet networks or those with Ethernet-like characteristics
- Other **non-TCP/IP** protocol suites
 - High-Level Data Link Control (HDLC)
 - Frame relay
 - Asynchronous Transfer Mode (ATM)

TCP/IP Internet Layer

- Internet layer pack data into data packets known as IP datagrams, which contain source and destination address (logical address or IP address) information that is used to forward the datagrams between hosts and across networks.
- The Internet layer is also responsible for routing of IP datagrams.
- The main responsibility of the internet layer is to send the packets from any network, and they arrive at the destination irrespective of the route they take.

TCP/IP Internet Layer Functions

- Protocols handle routing between machines across multiple networks
- Contains all functionality that manages the movement of data between two network devices over a routed network.
- is responsible for addressing, packaging, and routing functions.
- Three primary tasks
 - **MTU fragmentation**
 - **Addressing**
 - **Routing**

TCP/IP Internet Layer Protocols

- Protocols include
 - Internet Protocol (IP)
 - Internet Control Message Protocol (ICMP)
 - Packet Internetwork Groper (PING)
 - Address Resolution Protocol (ARP)
 - Reverse ARP (RARP)
 - Bootstrap Protocol (BOOTP)
 - Routing Information Protocol (RIP)
 - Open Shortest Path First (OSPF)
 - Border Gateway Protocol (BGP)

TCP/IP Transport Layer Functions

- The purpose of **Transport layer** is to permit devices **on the source and destination hosts** to carry on a conversation.
- **Transport layer** defines the **level of service** and **status of the connection** used when transporting data.
- The transport layer is responsible for the **reliability, flow control, and correction of data** which is being sent over the network.
- **Functions**
 - Reliable delivery of data from sender to receiver
 - Segmentation of outgoing messages and their reassembly prior to delivery to the Application layer
- **Hosts**
 - Devices that operate on the Internet

Two TCP/IP Transport layer protocols

- Two TCP/IP Transport layer protocols
 - Transmission Control Protocol (TCP)
 - **Connection-oriented**
 - User Datagram Protocol (UDP)
 - **Connectionless** UDP is often described as **connectionless** or “best-effort delivery” because it does not establish a connection before sending, it does not sequence packets before sending, and it does not provide error control through retransmission.
- User Datagram Protocol (UDP)
 - provides **connectionless** service and **end-to-end delivery** of transmission.
 - It is an **unreliable** protocol as it discovers the errors but not specify the error, and ICMP protocol reports the error to the sender that user datagram has been damaged.
 - **UDP** provides a **one-to-one or one-to-many, connectionless, unreliable communications service**.
- UDP
 - Transmits data in a “best-effort delivery”
 - Does no follow-up checking on its receipt

Two TCP/IP Transport layer protocols

- **Transmission Control Protocol(TCP)**

- provides a **one-to-one, connection-oriented, reliable** communications service.
- It provides a full transport layer services to applications.
- It creates a **virtual circuit** between the sender and receiver, and it is active for the duration of the transmission.
- TCP is a **reliable** protocol as it detects the error and retransmits the damaged frames.
- Therefore, it ensures all the segments must be received and acknowledged before the transmission is considered to be completed and **a virtual circuit is discarded**.
- At the sending end, TCP divides the whole message into smaller units known as **segment**, and each segment contains a **sequence number** which is required for reordering the frames to form an original message.
- At the receiving end, TCP collects all the segments and reorders them based on sequence numbers.

TCP/IP Application Layer

- Also known as the *Process layer*
- the top most layer of four layer TCP/IP model.
- defines TCP/IP application protocols and how host programs interface with Transport layer services to use the network.
- Application layer includes all the higher-level protocols like DNS (Domain Naming System), **HTTP (Hypertext Transfer Protocol)**, Telnet, **SSH**, **FTP** (File Transfer Protocol), **TFTP** (Trivial File Transfer Protocol), **SNMP** (Simple Network Management Protocol), **SMTP (Simple Mail Transfer Protocol)**, **DHCP (Dynamic Host Configuration Protocol)**, RDP (Remote Desktop Protocol) etc.

Application Layer Protocol

- **HTTP** : is used to transfer files that make up the Web pages of the World Wide Web.
 - This protocol allows us to access the data over the world wide web. It transfers the data in the form of plain text, audio, video. It is known as a Hypertext transfer protocol as it has the efficiency to use in a hypertext environment where there are rapid jumps from one document to another.
- **SMTP**: is used for the transfer of mail messages and attachments.
 - The TCP/IP protocol that supports the e-mail is known as a Simple mail transfer protocol. This protocol is used to send the data to another e-mail address.
- **FTP**: is used for interactive file transfer
 - FTP is a standard internet protocol used for transmitting the files from one computer to another computer.

Application Layer Protocol

- **SNMP**: SNMP stands for Simple Network Management Protocol.
 - is a framework used for managing the devices on the internet by using the TCP/IP protocol suite.
 - is used between a network management console and network devices (routers, bridges, intelligent hubs) to collect and exchange network management information.
- **DNS**: DNS stands for Domain Name System.
 - is used to resolve a host name to an IP address.
 - An IP address is used to identify the connection of a host to the internet uniquely. But, people prefer to use the names instead of addresses. Therefore, the system that maps the name to the address is known as Domain Name System.
- **TELNET**: It is an abbreviation for Terminal Network.
 - It establishes the connection between the local computer and remote computer in such a way that the local terminal appears to be a terminal at the remote system.

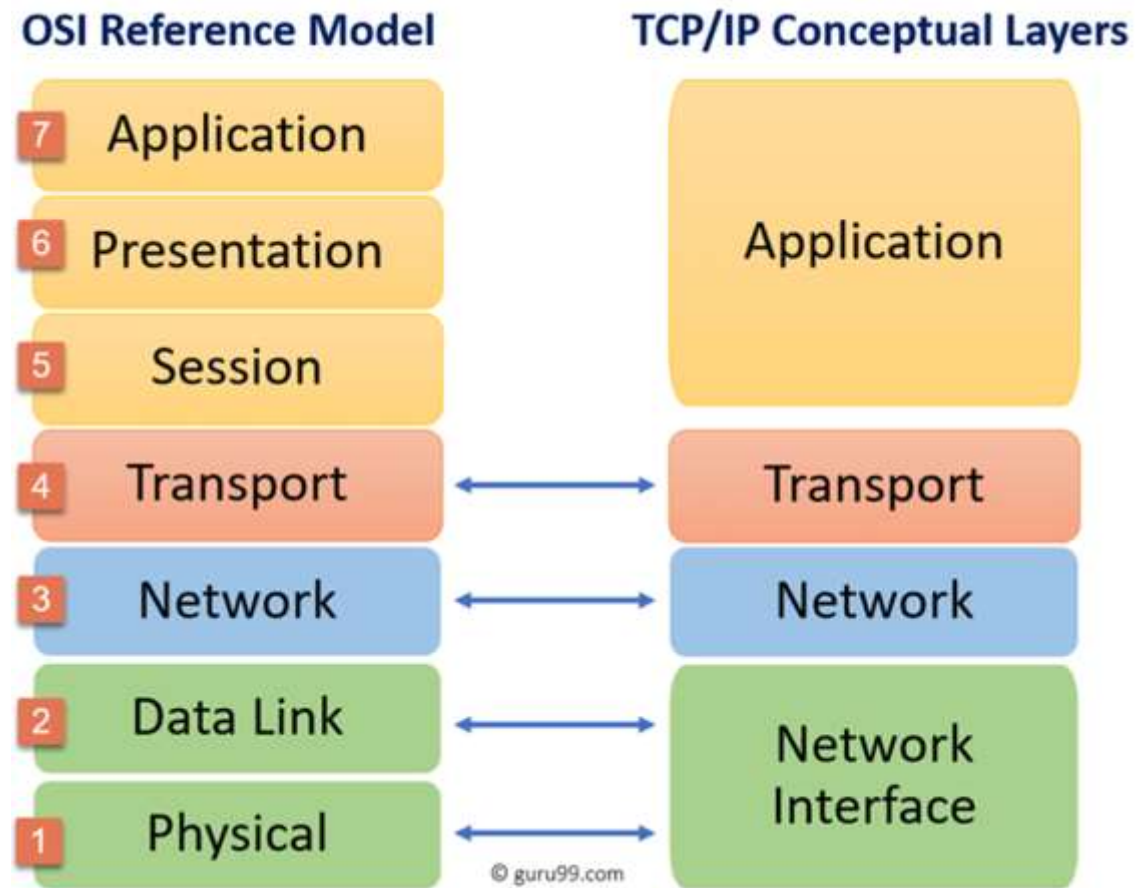
TCP/IP protocol suite

- **Application layer**
 - BGP, DHCP, DNS, FTP, HTTP, IMAP, LDAP, MGCP, NNTP, NTP, POP, ONC/RPC, RTP, RTSP, RIP, SIP, SMTP, SNMP, SSH, Telnet, TLS/SSL, XMPP
- **Transport layer**
 - TCP, UDP, DCCP, SCTP, RSVP
- **Internet layer**
 - IP, IPv4, IPv6, ICMP, ICMPv6, ECN, IGMP, IPsec
- **Link layer**
 - ARP, NDP, OSPF, Tunnels, L2TP, PPP, MAC, Ethernet, DSL, ISDN, FDDI,

Data Encapsulation in TCP/IP

- At each layer in the TCP/IP protocol stack
 - Outgoing data is packaged and identified for delivery to the layer underneath
- Header (or packet header)
 - PDU's own particular opening component
 - Identifies the protocol in use, the sender, and the intended recipient
- Trailer (or packet trailer)
 - Provides data integrity checks for the payload

1.5 Comparison of OSI and TCP/IP model



TCP/IP follows a horizontal approach, while OSI follows a vertical approach. In TCP/IP, the protocols were developed first and then the model. In OSI, the model was developed first and then the protocols in each layer were developed.

OSI(Open System Interconnection)	TCP/IP(Transmission Control Protocol / Internet Protocol)
It is developed by ISO (International Standard Organization)	It is developed by ARPANET (Advanced Research Project Agency Network).
1. OSI is a generic, protocol independent standard, acting as a communication gateway between the network and end user.	1. TCP/IP model is based on standard protocols around which the Internet has developed. It is a communication protocol, which allows connection of hosts over a network.
2. In OSI model the transport layer guarantees the delivery of packets.	2. In TCP/IP model the transport layer does not guarantees delivery of packets. Still the TCP/IP model is more reliable.
3. Follows vertical approach.	3. Follows horizontal approach.
4. OSI model has a separate Presentation layer and Session layer.	4. TCP/IP does not have a separate Presentation layer or Session layer.
5. Transport Layer is Connection Oriented.	5. Transport Layer is both Connection Oriented and Connection less.
6. Network layer of OSI model provides both connection oriented and connectionless service.	6. The Network layer in TCP/IP model provides connectionless service.
7...	7...
8. Protocols are hidden in OSI model and are easily replaced as the technology changes.	8. In TCP/IP replacing protocol is not easy.
9. OSI model defines services, interfaces and protocols very clearly and makes clear distinction between them. It is protocol independent.	9. In TCP/IP, services, interfaces and protocols are not clearly separated. It is also protocol dependent.
10. It has 7 layers	10. It has 4 layers

KEY DIFFERENCE

- OSI has 7 layers whereas TCP/IP has 4 layers.
- The OSI Model is a logical and conceptual model that defines network communication used by systems open to interconnection and communication with other systems. On the other hand, TCP/IP helps you to determine how a specific computer should be connected to the internet and how you can be transmitted between them.
- OSI header is 5 bytes whereas TCP/IP header size is 20 bytes.
- OSI refers to Open Systems Interconnection whereas TCP/IP refers to Transmission Control Protocol.
- OSI follows a vertical approach whereas TCP/IP follows a horizontal approach.
- OSI model, the transport layer, is only connection-oriented whereas the TCP/IP model is both connection-oriented and connectionless.
- OSI model is developed by ISO (International Standard Organization), whereas TCP Model is developed by ARPANET (Advanced Research Project Agency Network).
- OSI model helps you to standardize router, switch, motherboard, and other hardware whereas TCP/IP helps you to establish a connection between different types of computers.

End

- Assignment:
- Dear students, please prepare your copy accordingly.

1.6 Example network: The Internet, X.25, Frame Relay, Ethernet, VoIP, NGN and MPLS, xDSL.

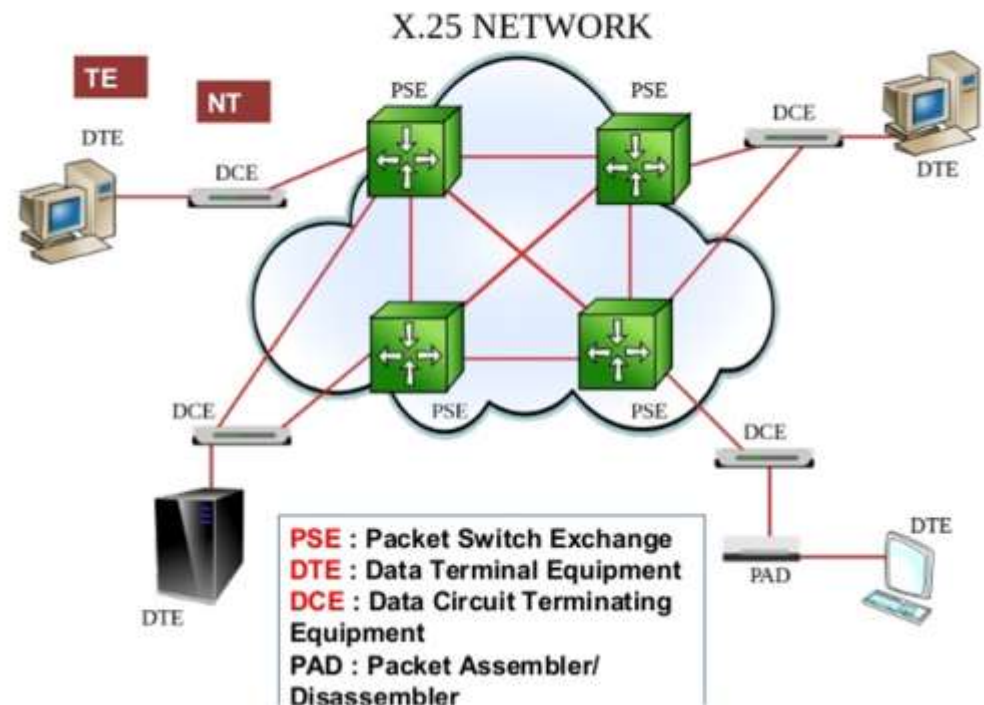
- **X.25**
- In the early 1970's there were many data communication networks (also known as Public Networks), which were owned by private companies, organizations and governments agencies.
- Since those public networks were quite different internally, and the interconnection of networks was growing very fast, there was a need for a common network interface protocol.
- In 1976, X.25 was recommended as the desired protocol by the International Consultative Committee for Telegraphy and Telephony (CCITT) called the International Telecommunication Union (ITU) since 1993.

X.25 Protocols

- X.25 was a standard suite of protocols used for packet-switched communications over a wide area network — a WAN, developed by ITU-T in 1976.
 - A **protocol** is an agreed-upon set of procedures and rules.
 - Two devices that follow the same protocols can understand each other and exchange data.
- X.25 is a standard for WAN communications that defines how connections between user devices and network devices are established and maintained.
- X.25 is designed to operate effectively regardless of the type of systems connected to the network.
- It is typically used in the packet-switched networks (PSNs) of common carriers, such as the telephone companies. Subscribers are charged based on their use of the network.
- The X.25 protocols works at the physical, data-link and Internet layer of the OSI model.

X.25 Devices and Protocol Operation

- X.25 network devices fall into three general categories:
- data terminal equipment (DTE),
- data circuit-terminating equipment (DCE),
- packet-switching exchange (PSE)



DTE/DCE/PSE

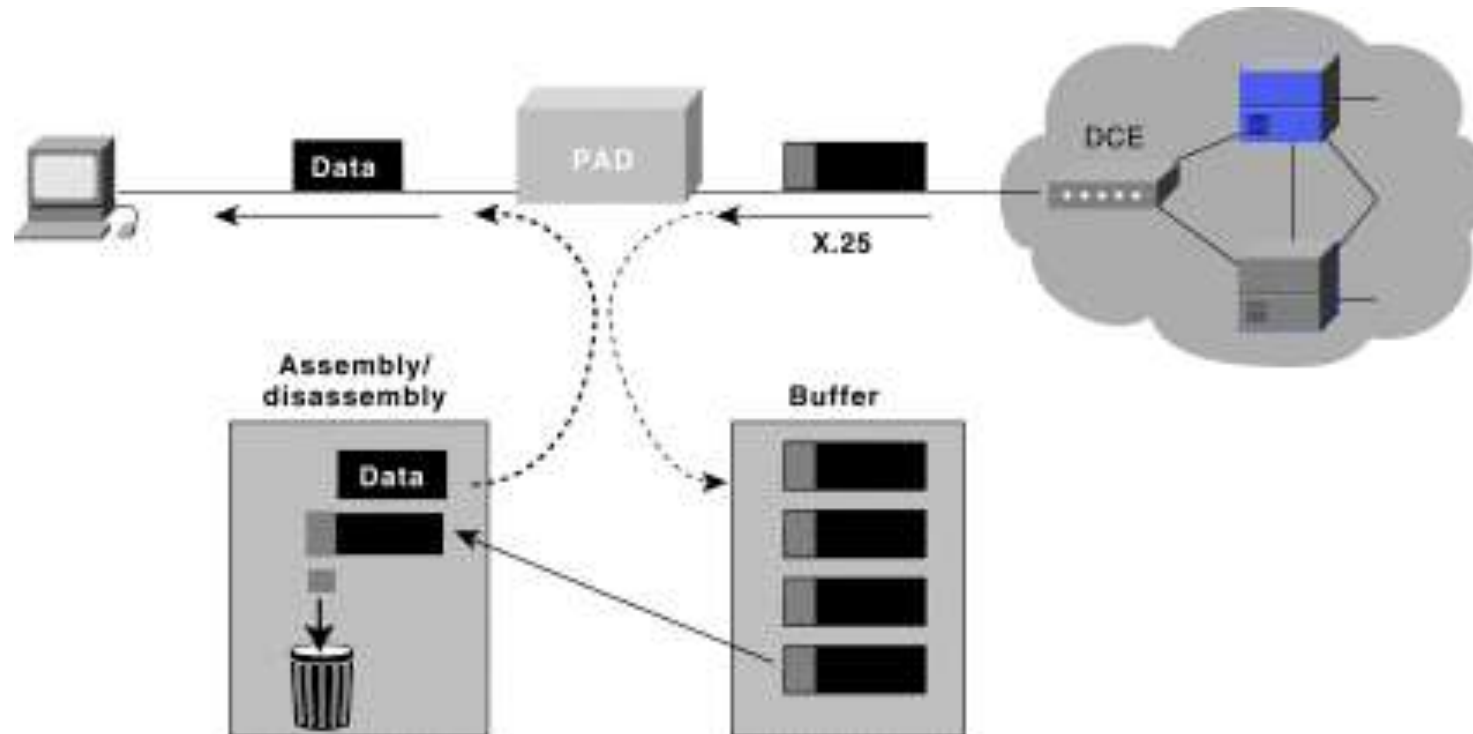
- **DTE** devices are end systems that communicate across the X.25 network.
 - They are usually terminals, personal computers, or network hosts, and are located on the premises of individual subscribers.
- **DCEs** are communications devices, such as modems and packet switches that provide the interface between DTE devices and a PSE, and are generally located in the carrier's facilities.
- **PSEs** are switches that compose the bulk of the carrier's network. They transfer data from one DTE device to another through the X.25 PSN.

Packet Assembler/Disassembler(PAD)

- PAD is a device commonly found in X.25 networks.
- PADs are used when a DTE device, such as a character-mode terminal, is too simple to implement the full X.25 functionality.
- The PAD is located between a DTE device and a DCE device, and it performs three primary functions:
 - buffering (storing data until a device is ready to process it),
 - packet assembly, and
 - packet disassembly.
- The PAD buffers data sent to or from the DTE device.
- It also assembles outgoing data into packets and forwards them to the DCE device.
 - (This includes adding an X.25 header.)
- Finally, the PAD disassembles incoming packets before forwarding the data to the DTE.
 - (This includes removing the X.25 header)

PADs

- Figure illustrates the basic operation of the PAD when receiving packets from the X.25 WAN. (disassembling)



Session Establishment

- X.25 sessions are established when one DTE device contacts another to request a communication session.
- It's up to the receiving DTE whether to accept or refuse the connection.
- If the request is accepted, the two systems begin full-duplex communication.
- Either DTE device can terminate the connection.
- After the session is terminated, any further communication requires the establishment of a new session.

Is x.25 still used?

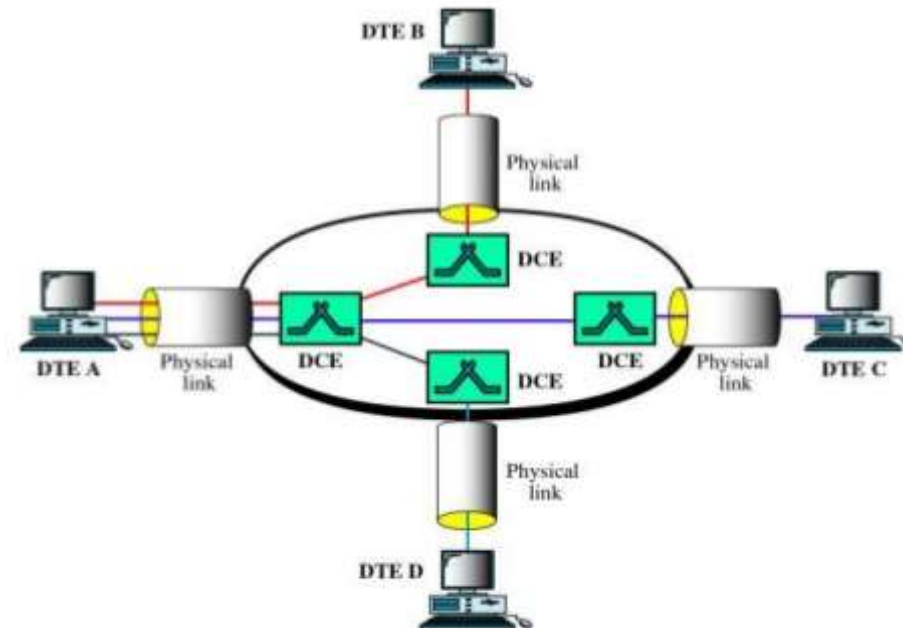
- **X.25** is an ITU-T standard protocol suite for packet-switched data communication in wide area networks (WAN). ... However, most users have moved to Internet Protocol (IP) systems instead.
- **X. 25** was **used** up to 2015 (e.g. by the credit card payment industry) and is **still used** by aviation, purchasable from telecoms companies.

Virtual Circuits

- The X.25 is a **packet-switched** virtual circuit network.
- A virtual circuit is a logical connection created to ensure reliable communication between two network devices.
- A virtual circuit denotes the existence of a logical, bidirectional path from one DTE device to another across an X.25 network.
- Physically, the connection can pass through any number of intermediate nodes, such as DCE devices and PSEs.
- Virtual circuits in X.25 are created at the network layer such that multiple virtual circuits (logical connections) can be multiplexed onto a single physical circuit (a physical connection).
- Virtual circuits are demultiplexed at the remote end, and data is sent to the appropriate destinations.

...

Virtual Circuits in X.25



In X.25, SVCs and PVCs both are used.

5

Virtual Circuits...

- X.25 permits a DTE user on an **X.25 network** to communicate with a number of remote DTE's simultaneously. Connections occur on logical channels of two types:
- **Switched virtual circuits (SVCs)** – SVCs are very much like telephone calls; a connection is established, data are transferred and then the connection is released. Each DTE on the network is given a unique DTE address which can be used much like a telephone number.
- SVCs are temporary connections used for sporadic data transfers. They require that two DTE devices to establish, maintain, and terminate a session each time the devices need to communicate.

Virtual Circuits...

- **Permanent virtual circuits (PVCs)** – a PVC is similar to a leased line in that the connection is always present. The logical connection is established permanently by the Packet Switched Network administration. Therefore, data may always be sent, without any call setup.
- PVCs are permanently established connections used for frequent and consistent data transfers. PVCs do not require that sessions be established and terminated. Therefore, DTEs can begin transferring data whenever necessary because the session is always active.

Operation of an X.25 virtual circuit

- The basic operation of an X.25 virtual circuit begins when the source DTE device specifies the virtual circuit to be used (in the packet headers) and then sends the packets to a locally connected DCE device.
- At this point, the local DCE device examines the packet headers to determine which virtual circuit to use and then sends the packets to the closest PSE in the path of that virtual circuit.
- PSEs (switches) pass the traffic to the next intermediate node in the path, which may be another switch or the remote DCE device.
- When the traffic arrives at the remote DCE device, the packet headers are examined and the destination address is determined.
- The packets are then sent to the destination DTE device.
- If communication occurs over an SVC and neither device has additional data to transfer, the virtual circuit is terminated.

X.25 Protocol Suite

- The X.25 protocol suite maps to [the lowest three layers of the OSI reference model](#).
- **Physical layer:** Deals with the physical interface between an attached station and the link that attaches that station to the packet-switching node.
 - X.21 is the most commonly used physical layer standard.
- **Frame layer:** Facilitates reliable transfer of data across the physical link by transmitting the data as a sequence of frames.
 - Uses a subset of HDLC known as **Link Access Protocol Balanced (LAPB)**, bit oriented protocol.
- **Packet layer:** Responsible for end-to-end connection between two DTEs. Functions performed are:
 - Establishing connection
 - Transferring data
 - Terminating a connection
 - Error and flow control
 - With the help of X.25 packet layer, data are transmitted in packets over external virtual circuits.

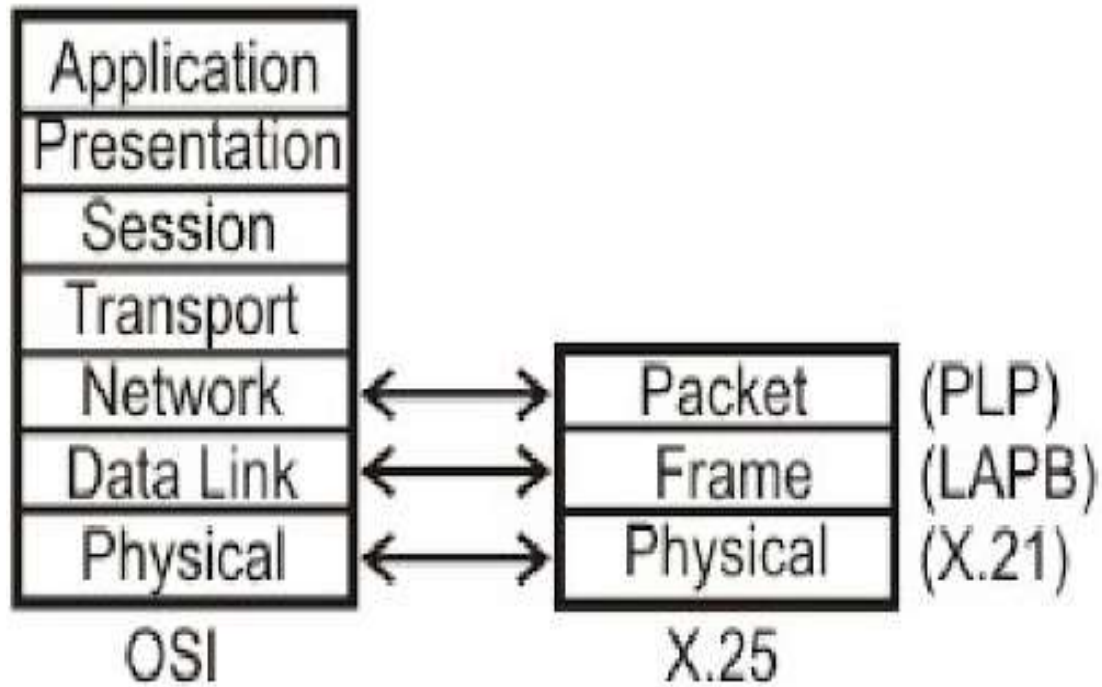


Figure 4.4.4 X.25 Layer mapping with OSI model

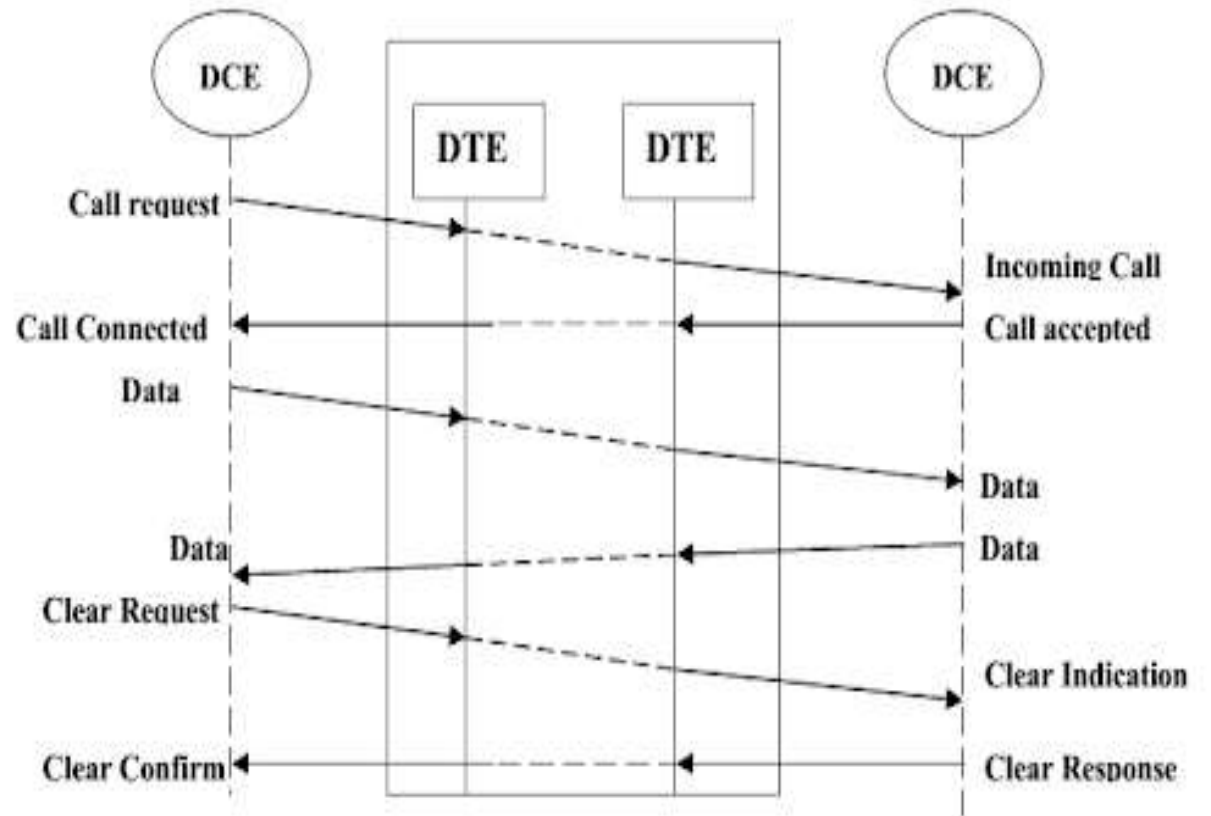


Figure 4.4.6 Different Modes of PLP

5 PLP modes

- Packet Layer Protocol (PLP) is the network layer protocol of X.25. The PLP operates in five distinct modes
- **Call setup mode** is used to establish SVCs between DTE devices.
- A PLP uses the X.121 addressing scheme to set up the virtual circuit.
- The call setup mode is executed on a per virtual-circuit basis, which means that one virtual circuit can be in call setup mode while another is in data transfer mode.
- This mode is used **only with SVCs, not with PVCs**.
- To establish a connection on an SVC, the calling DTE sends a Call Request Packet, which includes the address of the remote DTE to be contacted.
- The destination DTE decides whether or not to accept the call (the Call Request packet includes the sender's DTE address, as well as other information that the called DTE can use to decide whether or not to accept the call).
- A call is accepted by issuing a Call Accepted packet, or cleared by issuing a Clear Request packet.
- Once the originating DTE receives the Call Accepted packet, the virtual circuit is established and data transfer may take place.

5 PLP modes...

- **Data transfer mode** is used for transferring data between two DTE devices across a virtual circuit. In this mode, PLP handles segmentation and reassembly, bit padding, and error and flow control. This mode is executed **on a per-virtual-circuit** basis and is used **with both PVCs and SVCs**.
- **Idle mode** is used when a virtual circuit is established but data transfer is not occurring. It is executed **on a per-virtual-circuit** basis and is used **only with SVCs**.
- **Call clearing mode** is used to end communication sessions between DTE devices and to terminate SVCs. This mode is executed **on a per-virtual-circuit** basis and is used **only with SVCs**. When either DTE wishes to terminate the call, a Clear Request packet is sent to the remote DTE, which responds with a Clear Confirmation packet.
- **Restarting mode** is used to synchronize transmission between a DTE device and a locally connected DCE device. This mode is **not** executed **on a per-virtual circuit** basis. It affects all the DTE device's established virtual circuits.

Frame Relay

- Frame Relay is a high-performance WAN protocol that operates at the **physical and data link layers** of the OSI reference model.
- Frame Relay originally was designed for use across Integrated Services Digital Network (ISDN) interfaces.
 - Today, it is used over a variety of other network interfaces as well.
- Frame Relay is a simplified form of Packet Switching, similar in principle to X.25, in which synchronous frames of data are routed to different destinations depending on header information.
- The biggest difference between Frame Relay and X.25 is that X.25 guarantees **data integrity** and **network managed flow control** at the cost of some network delays.
- Frame Relay switches packets end to end much faster, but there is **no** guarantee of data integrity at all.

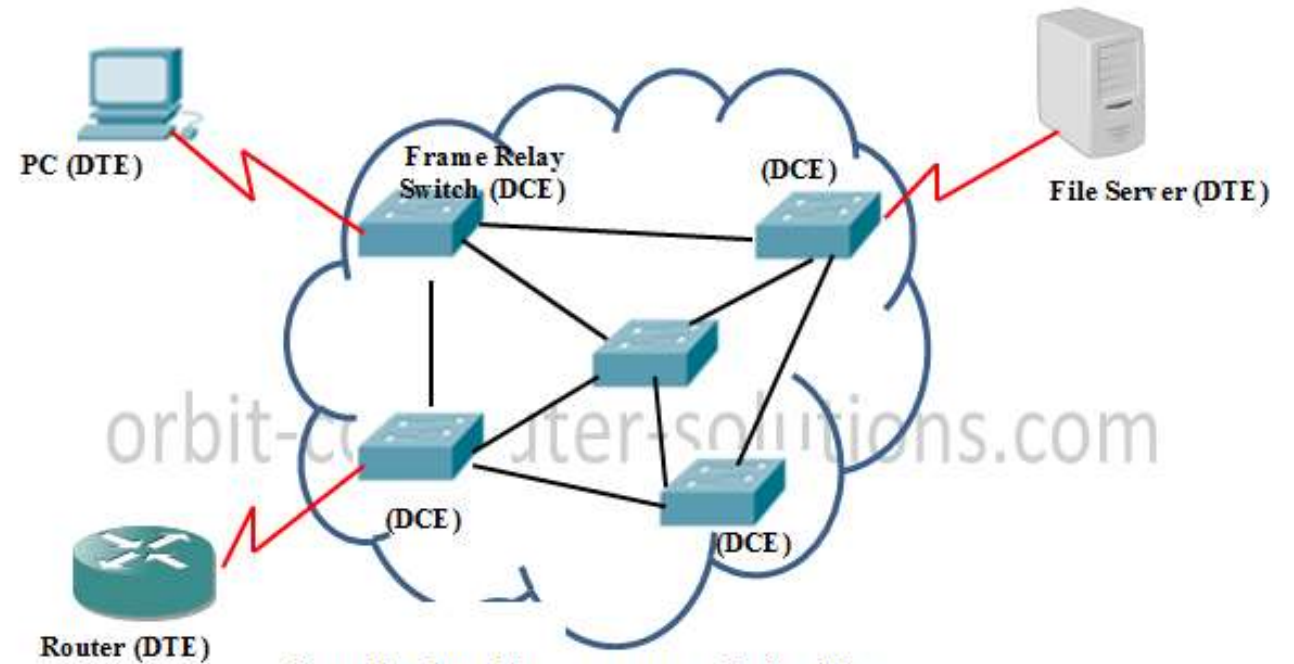
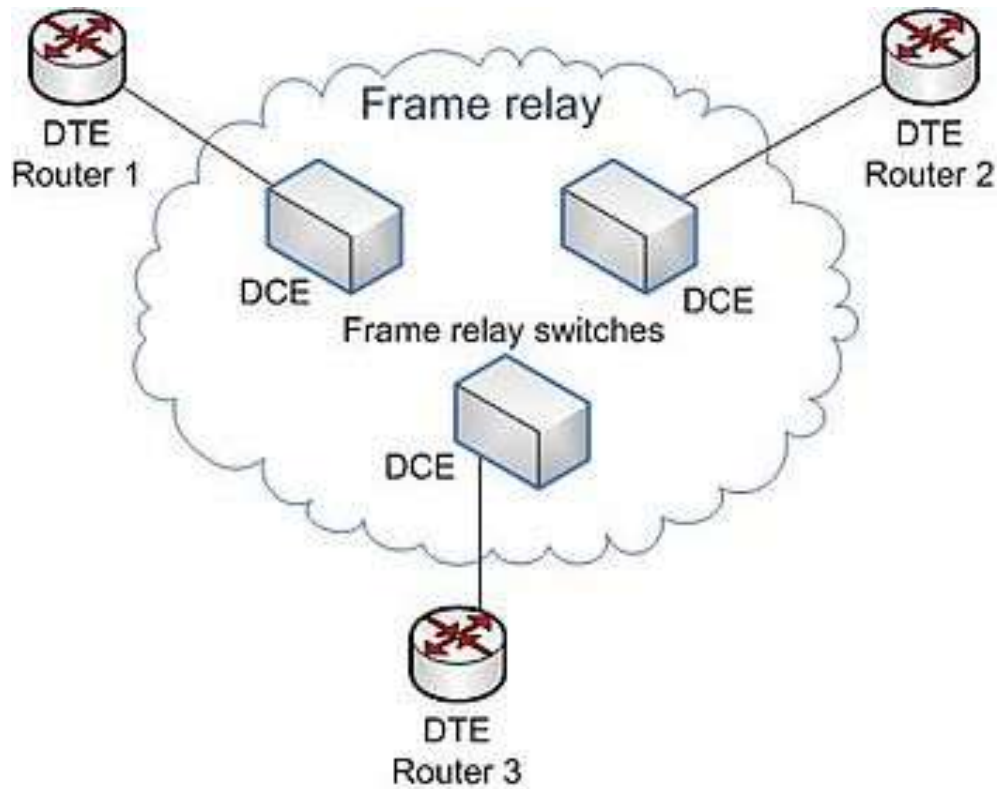
Frame Relay Devices

- Frame Relay is an example of a **packet-switched technology**
- Devices attached to a Frame Relay WAN fall into the following two general categories:
 - Data terminal equipment (DTE)
 - DTEs generally are considered to be terminating equipment for a specific network and typically are located on the premises of a customer.
 - In fact, they may be owned by the customer.
 - Examples of DTE devices are **terminals, personal computers, routers, and bridges**.
 - Data circuit-terminating equipment (DCE)
 - DCEs are carrier-owned internetworking devices.
 - The purpose of DCE equipment is to provide clocking and switching services in a network, which are the devices that actually transmit data through the WAN.
 - In most cases, these are **packet switches**.

Frame Relay Devices...

- The connection between a DTE device and a DCE device consists of both a **physical layer** component and a **link layer** component.
- The physical component defines the mechanical, electrical, functional, and procedural specifications for the connection between the devices.
 - One of the most commonly used physical layer interface specifications is the recommended standard (RS)-232 specification.
- The link layer component defines the protocol that establishes the connection between the DTE device, such as a router, and the DCE device, such as a switch.

Frame Relay



From the above figure, you see mesh of switches interconnected by trunk links. Frame Relay simply determines how data moves between DTEs and DCEs over a local loop.

Virtual Circuits

- Frame Relay is a virtual circuit network, so it doesn't use physical addresses to define the DTEs connected to the network.
- Frame Relay provides connection-oriented data link layer communication. This means that a defined communication exists between each pair of devices and that these connections are associated with a connection identifier.
- However, virtual circuit identifiers in Frame relay operate at the data link layer, in contrast with X.25, where they operate at the network layer.
- This service is implemented by using a Frame Relay virtual circuit, which is a logical connection created between two data terminal equipment (DTE) devices across a Frame Relay packet-switched network (PSN).
- Virtual circuits provide a bidirectional communication path from one DTE device to another and are uniquely identified by a data-link connection identifier (DLCI).

Virtual Circuits...

- A number of virtual circuits can be multiplexed into a single physical circuit for transmission across the network.
- This capability often can reduce the equipment and network complexity required to connect multiple DTE devices.
- A virtual circuit can pass through any number of intermediate DCE devices (switches) located within the Frame Relay PSN.
- Two types of Frame Relay Circuits:
 - switched virtual circuits (SVCs) and
 - permanent virtual circuits (PVCs).

SVCs

- SVCs are temporary connections used in situations requiring only sporadic data transfer between DTE devices across the Frame Relay network.
- A communication session across an SVC consists of the following four operational states:
- **Call setup**—The virtual circuit between two Frame Relay DTE devices is established.
- **Data transfer**—Data is transmitted between the DTE devices over the virtual circuit.
- **Idle**—The connection between DTE devices is still active, but no data is transferred. If an SVC remains in an idle state for a defined period of time, the call can be terminated.
- **Call termination**—The virtual circuit between DTE devices is terminated. After the virtual circuit is terminated, the DTE devices must establish a new SVC if there is additional data to be exchanged.
 - It is expected that SVCs will be established, maintained, and terminated using the same signaling protocols used in ISDN.

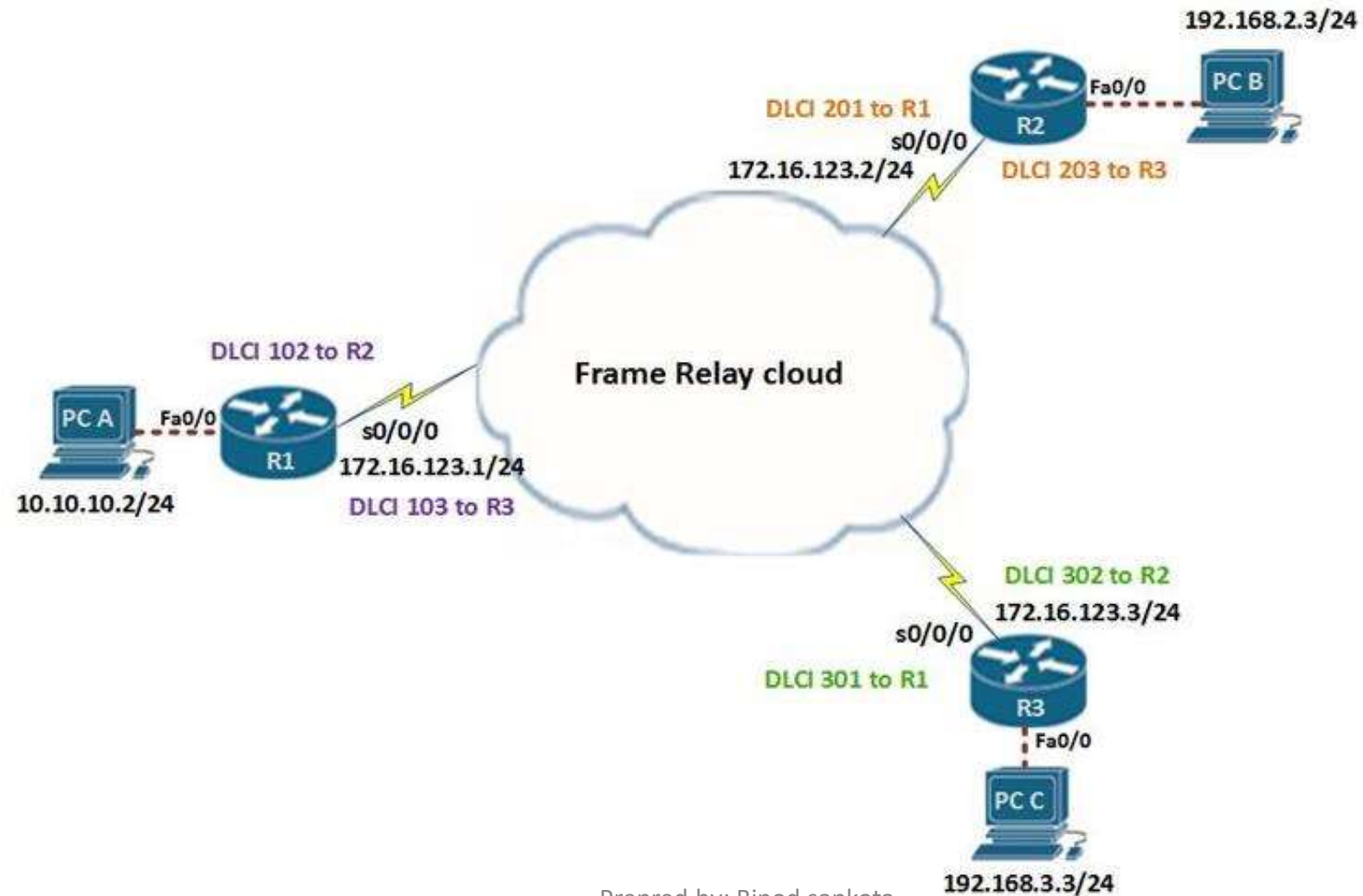
PVCs

- PVCs are permanently established connections that are used for frequent and consistent data transfers between DTE devices across the Frame Relay network.
- Communication across PVC does not require the call setup and termination states that are used with SVCs.
- PVCs always operate in one of the following two operational states:
- **Data transfer**: Data is transmitted between the DTE devices over the virtual circuit.
- **Idle**: The connection between DTE devices is active, but no data is transferred.
- Unlike SVCs, PVCs will not be terminated under any circumstances when in an idle state.
- DTE devices can begin transferring data whenever they are ready because the circuit is permanently established.

Data-Link Connection Identifier (DLCI)

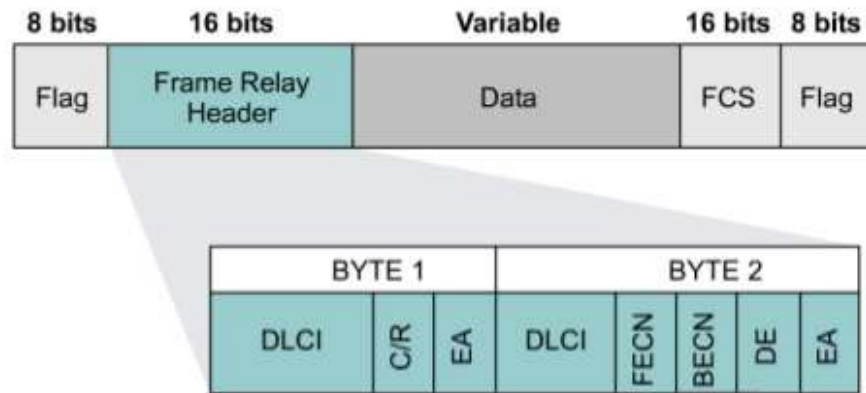
- Frame Relay virtual circuits are identified by data-link connection identifiers (DLCIs).
- DLCI values typically are assigned by the Frame Relay service provider (for example, telephone company).
- Frame Relay DLCIs have local significance, which means that their values are unique in the LAN, but not necessarily in the Frame Relay WAN.
- The local DTEs use this DLCI to send frames to the remote DTE.

DLCI



IETF Frame Relay Frame

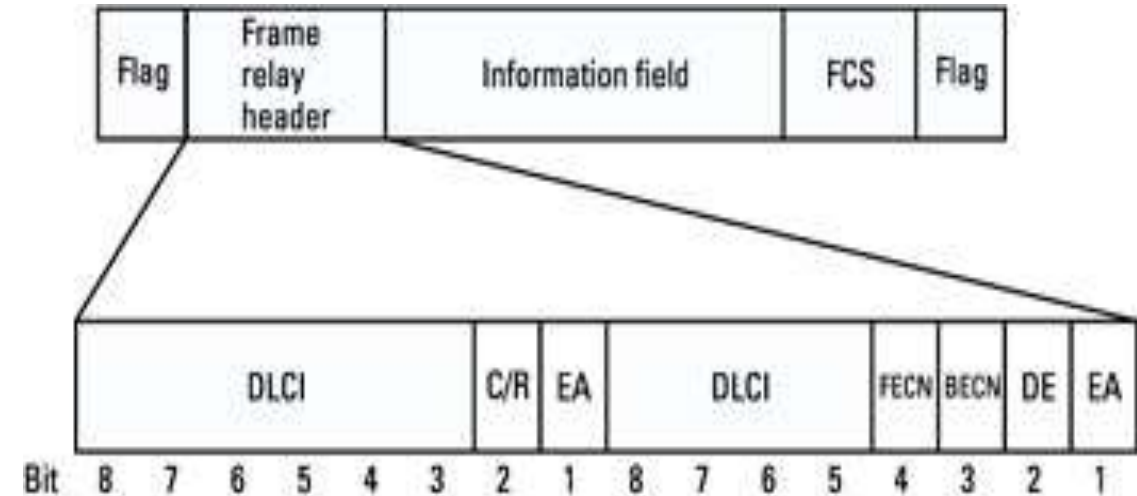
Cabrillo College



- Cisco routers support two types of Frame Relay headers.
 - **Cisco**, which is a 4-byte header.
 - **IETF**, which is a 2-byte header that conforms to the IETF standards.
- The Cisco proprietary 4-byte header is the default and cannot be used if the router is connected to another vendor's equipment across a Frame Relay network.

Rich Grier, rgrier@cabrillo.edu

12



DLCI: Data link connection identifier
 C/R: Command/response field bit
 FECN: Forward explicit congestion notification
 BECN: Backward explicit congestion notification
 DE: Discard eligibility indicator
 EA: Extension bit

Frame Relay Summary

- Frame relay operates only in data link and physical layer.
- Frame Relay allows bursty traffic.
- It allows frame size of **9000 bytes**, which can accommodate all local area network frames.
- Frame relay is less expensive than other traditional WANs.
- Frame relay provides both Permanent and switched connections.
- Frame relay allow variable-length frames, this may create varying delays for different users.
 - Due to variable delay it is not suitable for real-time communication



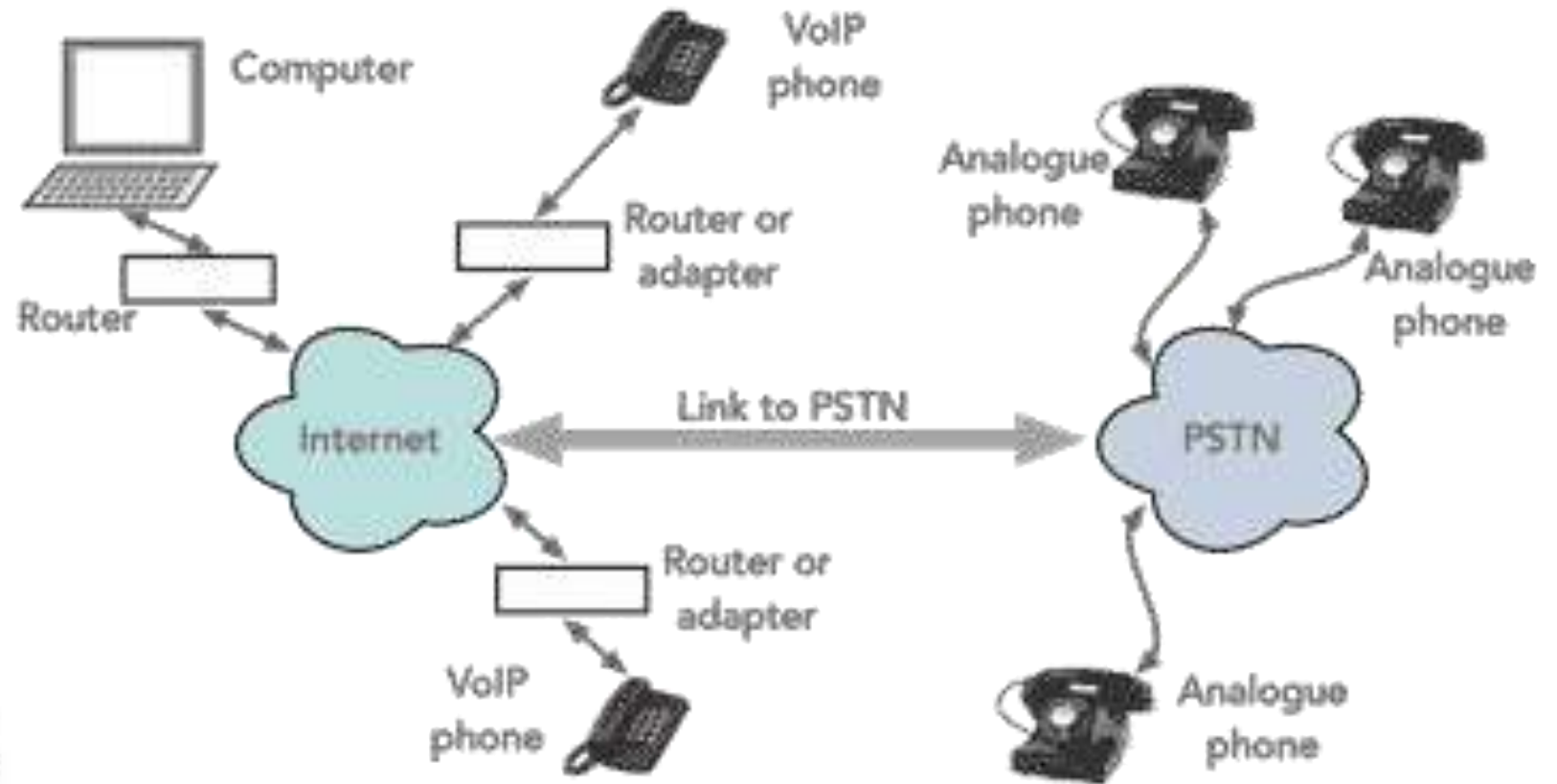
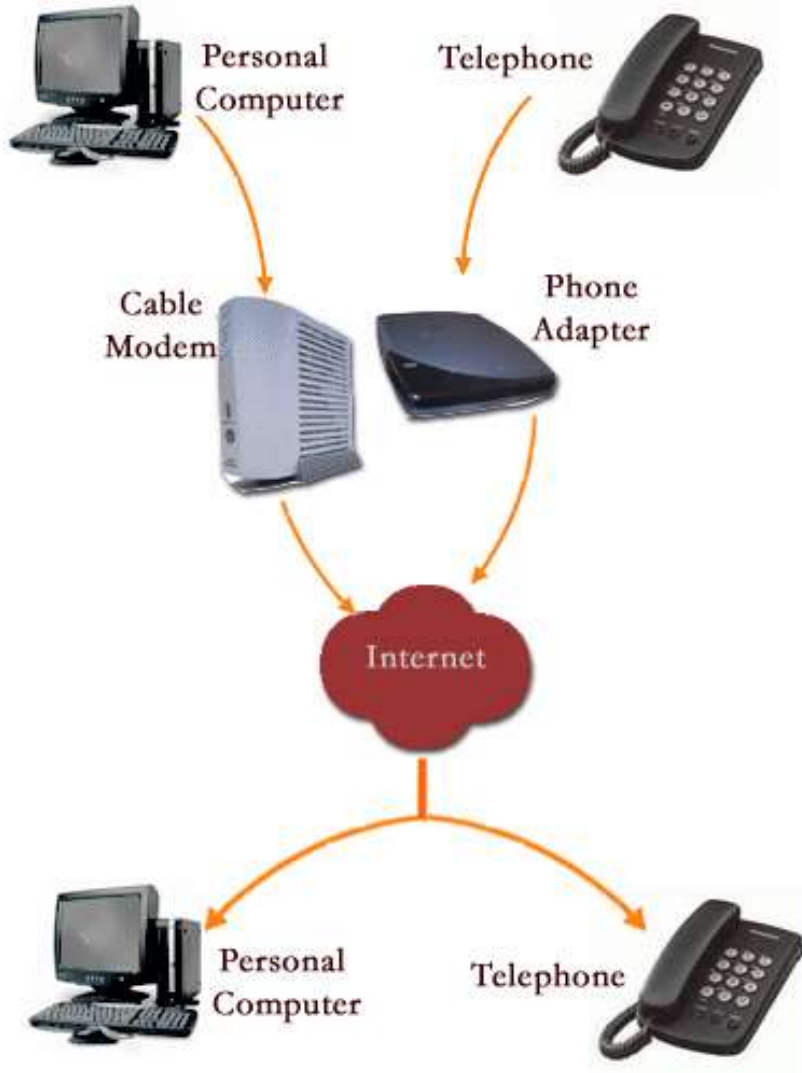
Frame Relay vs. X.25

Frame Relay	X.25
■ No error detection -> greater speeds	■ Error detection -> error-free delivery
■ Physical and data link layers. -> high performance, greater transmission	■ Physical, data link and network layers
■ Prepare and send frames ■ Frames contain expanded address field -> direct frames to destinations with minimal processing	■ Prepare and send packets ■ Packets contain fields used for error and flow control
■ Can dynamically allocate bandwidth	■ Has fixed bandwidth available

Voice over Internet Protocol (VoIP)

- Voice communication carried out using the IP for the transport is known as VoIP.
- VoIP services convert our voice into a digital signal that travels over the Internet.
- VoIP can allow us to make a call directly from a computer, a special VoIP phone, or a traditional phone connected to a special adapter.
- Requirements:
 - A broadband (high speed Internet) connection is required.
 - A computer, adaptor, or specialized phone is required.

VoIP



NGN

- The general idea - (voice, data, and all sorts of media such as video) by encapsulating these into IP packets, similar to those used on the Internet.
- Voice + data + multimedia -> Packets
- NGNs are commonly built around the Internet Protocol, and therefore the term all IP is also sometimes used to describe the transformation of formerly telephone-centric networks toward NGN.
- NGN is a body of key architectural changes in telecommunication core (which connects local providers to one another) and access (which connects subscribers to their immediate service provider) networks.

NGN

- According to [ITU-T](#),
- A next-generation network (NGN) is a packet-based network which can provide services including Telecommunication Services and is able to make use of multiple broadband, quality of Service-enabled transport technologies and in which service-related functions are independent from underlying transport-related technologies.
- A major goal of NGN is to facilitate convergence of networks and convergence of services.

Objectives of ITU-T NGN

- promote fair competition
- encourage private investment
- define a framework for architecture and capabilities to be able to meet various regulatory requirements
- provide open access to networks
- ensuring universal provision and access to services
- promoting equality of opportunity to the citizen
- promoting diversity of content, including cultural and linguistic diversity
- recognizing the necessity of worldwide cooperation with particular attention to less developed countries

NGN characteristics

- Packet-based transfer
- Separation of control functions among bearer capabilities, call/session, and application/ service
- Decoupling of service provision from network, and provision of open interfaces
- Support for a wide range of services, applications and mechanisms based on service building blocks (including real time/ streaming/ non-real time services and multi-media)
- Broadband capabilities with end-to-end QoS and transparency
- Interworking with legacy networks via open interfaces
- Generalized mobility

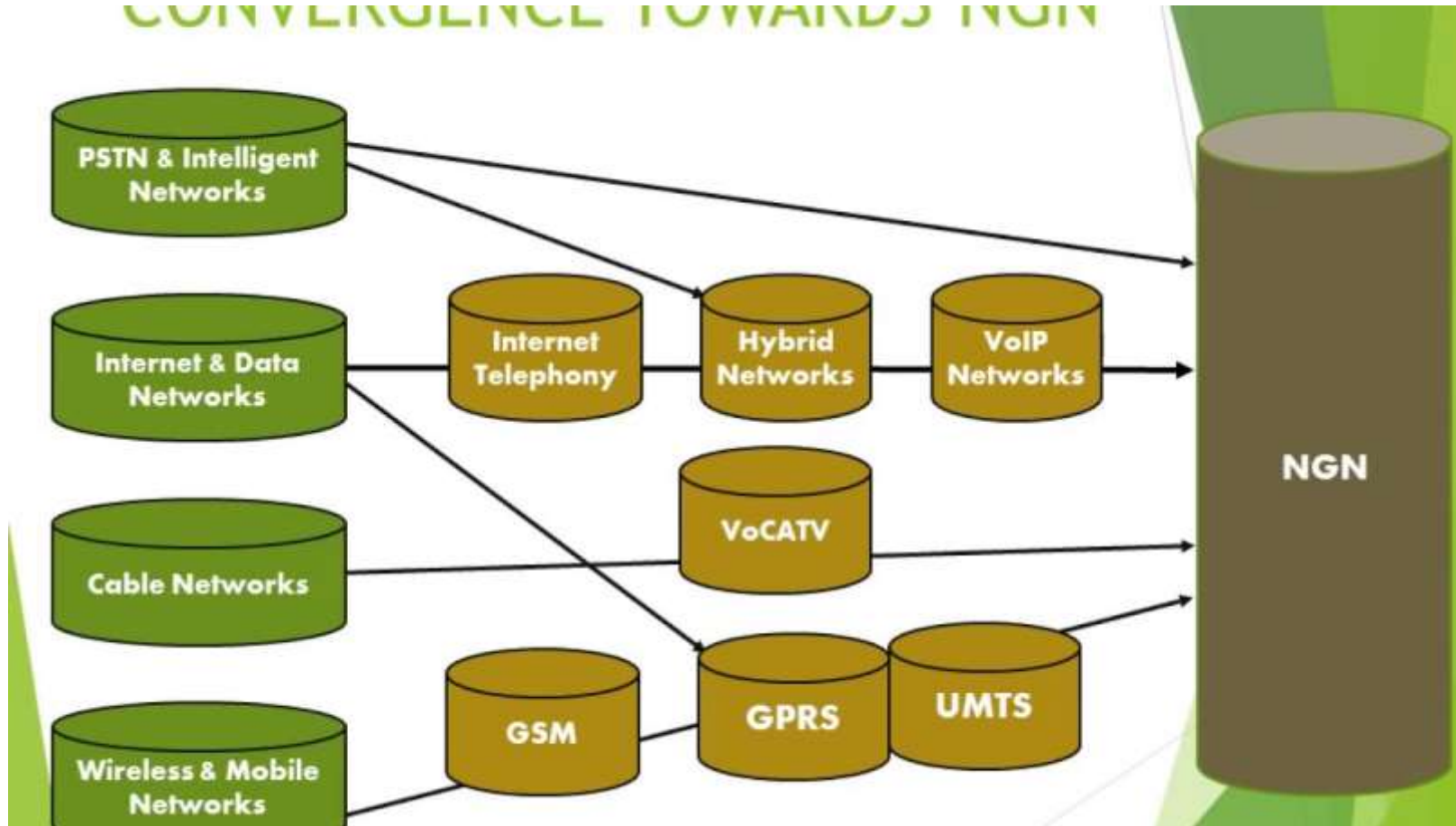
NGN characteristics...

- Unrestricted access by users to different service providers
- A variety of identification schemes which can be resolved to IP addresses for the purposes of routing in IP networks
- Unified service characteristics for the same service as perceived by the user
- Converged services between Fixed/Mobile
- Independence of service-related functions from underlying transport technologies
- Compliant with all Regulatory requirements, for example concerning emergency communications and security/privacy, etc.

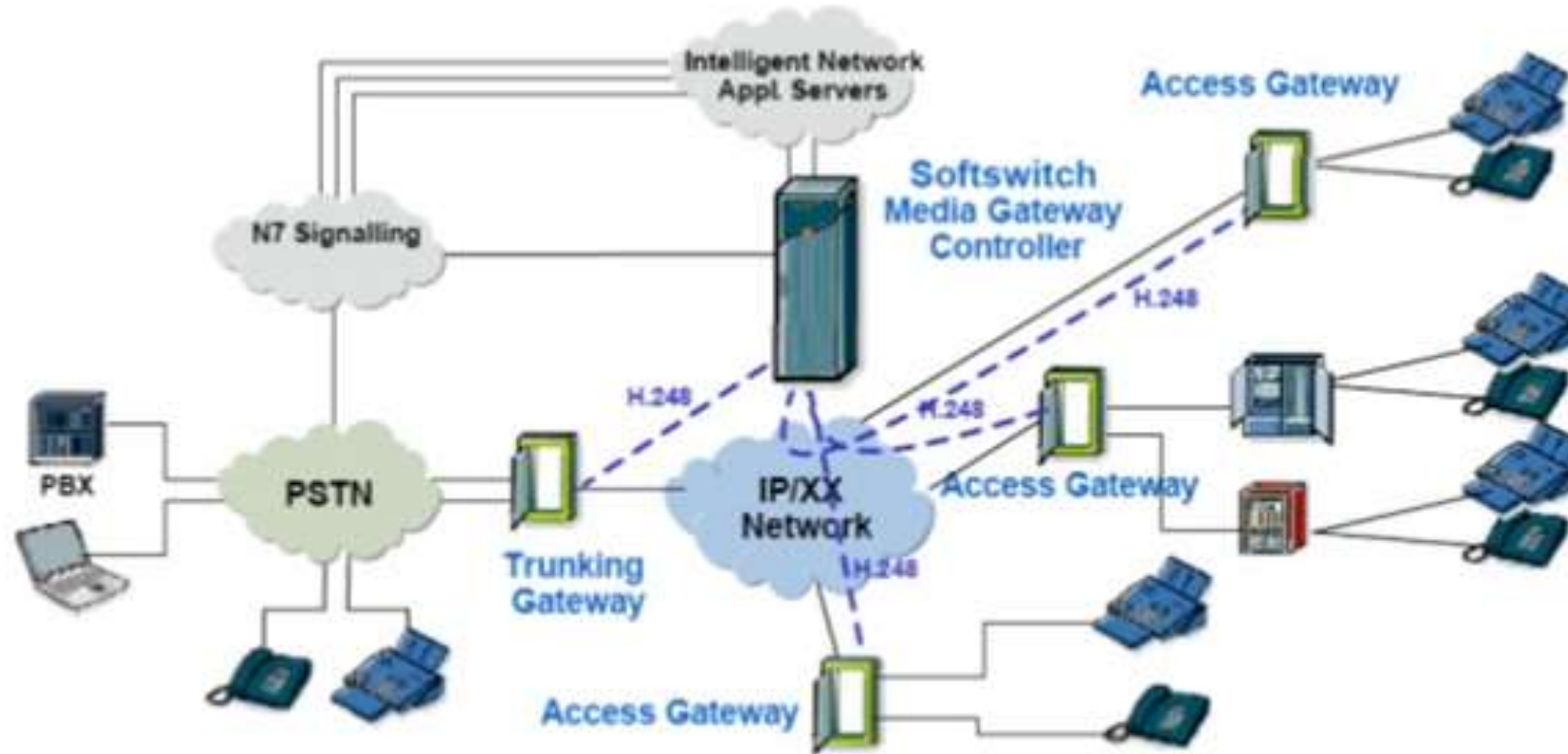
Convergence towards NGN

- Today's network is divided into
 - The public switched telephone network
 - The packet switched network (e.g. the internet)and
 - The mobile networks
- Convergence is the process of interconnection of traditional switched networks (The PSTN and mobile networks) and packet switched networks based on the internet protocol(IP) for routing.

Convergence towards NGN



NGN Elements



Source: ITU/BDT NGN Network Architecture

NGN Network Elements

- **Access Gateways**

- allows the connection of subscriber lines to the packet network
- converts the traffic flows of analogue signal to packet network
- provides subscriber access to NGN network and services

- **Trunking Gateways**

- allows interworking between classical TDM telephony network and Packet-based NGN networks,
- converts TDM circuits/ trunks (64kbps) flows into data packets, and vice versa

NGN Network Elements

- **Soft switch/MGC**
- referred to as the Call Agent or Media Gateway Controller (MGC).
- provides the “service delivery control” within the network
 - (Access and/or Trunking) via H.248 protocol
- performs signalling gateway functionality or uses a signalling gateway for interworking with PSTN N7 signalling network
- provides connection to Intelligent Network /applications servers (AS) to offer the same services as those available to TDM subscribers
- **Application server**
- Support service execution e.g. to control cal server and NGN special resources (e.g. media server, message server) via these protocol: H.248,SIP,SS7,ENUM,MPLS,BGP etc.

NGN Network Elements

- **H.248 Protocol**

- Known also as MEGACO: standard protocol, defined by ITU-T, for signalling and session management needed during a communication between a media gateway, and the media gateway controller managing it
- H.248/MEGACO allows to set up, keep, and terminate calls between multiple endpoints as between telephone subscribers using the TDM

- **SIP**

- Session Initiation Protocol in order to handle communication signalling and negotiation like call establishment, maintenance and termination from packet mode terminals. Has a distributed peer to peer implementation

NGN Network Elements

- **Signaling Gateway (SG)**
- A unit that provides signaling conversion between the NGN and the other networks (e.g. STP in SS7).
- **ENUM**
- Electronic NUMbering: Protocol that allows to establish a correspondence between the traditional telephone numbering (E.164) and the network addresses related to the packet mode networks (RFC 2916 "E.164 number and DNS" IETF).

NGN Network Elements

- **MPLS** : Multiprotocol Label Switch or protocol that assigns labels to information packets in order to allow the node routers to treat and route flows in the network paths according to established **priority** for each category.
- Establishes a tunnel for an end to end forwarding. A label is a short, fixed length, locally significant identifier which is used to identify a "Forwarding Equivalence Class" (FEC) to which that packet is assigned."
- **LSP**: Label-switched paths: An LSP is a specific traffic path that using convenient protocols will establish a path through an MPLS network and will reserve necessary resources to meet pre-defined service requirements for the data path.

NGN Network Elements

- **OSPF**

- Open Shortest Path First: A routing protocol that determines the best path for routing IP traffic over a TCP/IP network based on distance between nodes and several quality parameters. OSPF is an interior gateway protocol (IGP), which is designed to work within an autonomous system

- **BGP**

- Border Gateway Protocol: performs inter-domain routing in TCP/IP networks, handling routing between multiple autonomous domains. Routers use BGP to maintain a consistent view of the internetwork topology

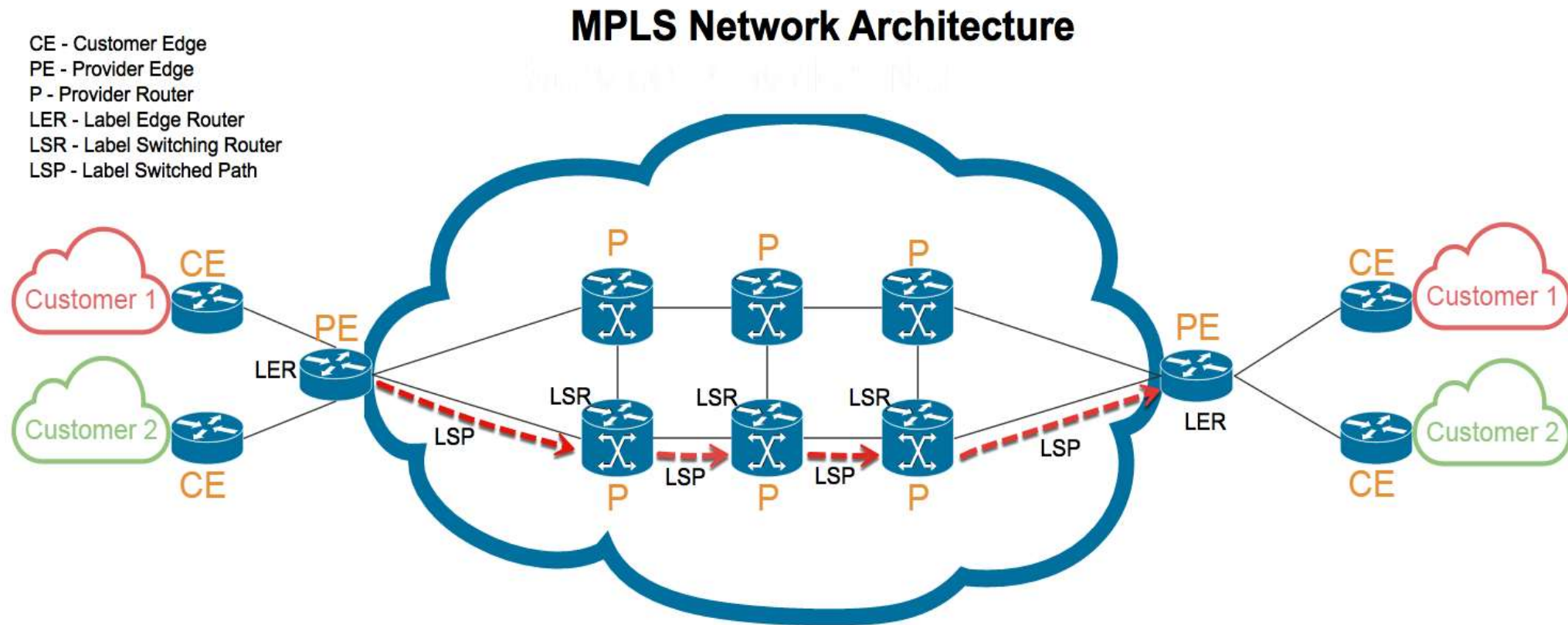
MPLS(Multi Protocol Label Switching)

- In case of a circuit switched network a fixed amount of bandwidth is reserved as well as a route is chosen before a connection is established and only this route is used for the complete transaction on the connection. **It is reliable but more expensive.**
- While in case of a packet switched network all the information is packaged into small groups called packets, frames, blocks, or cells.
- Circuit switching provides better Quality of Service (QOS) while packet switching utilizes the bandwidth more efficiently.
- Also, information about each and every connection needs to be stored thereby putting a limit on scalability.
- Packet switching is flexible and data path is not fixed, but **processing IP information at every single router slows down transmission.**
- **MPLS** tries to combine the benefit of both the above switching technologies.
- Defined by IETF in 1998.

MPLS

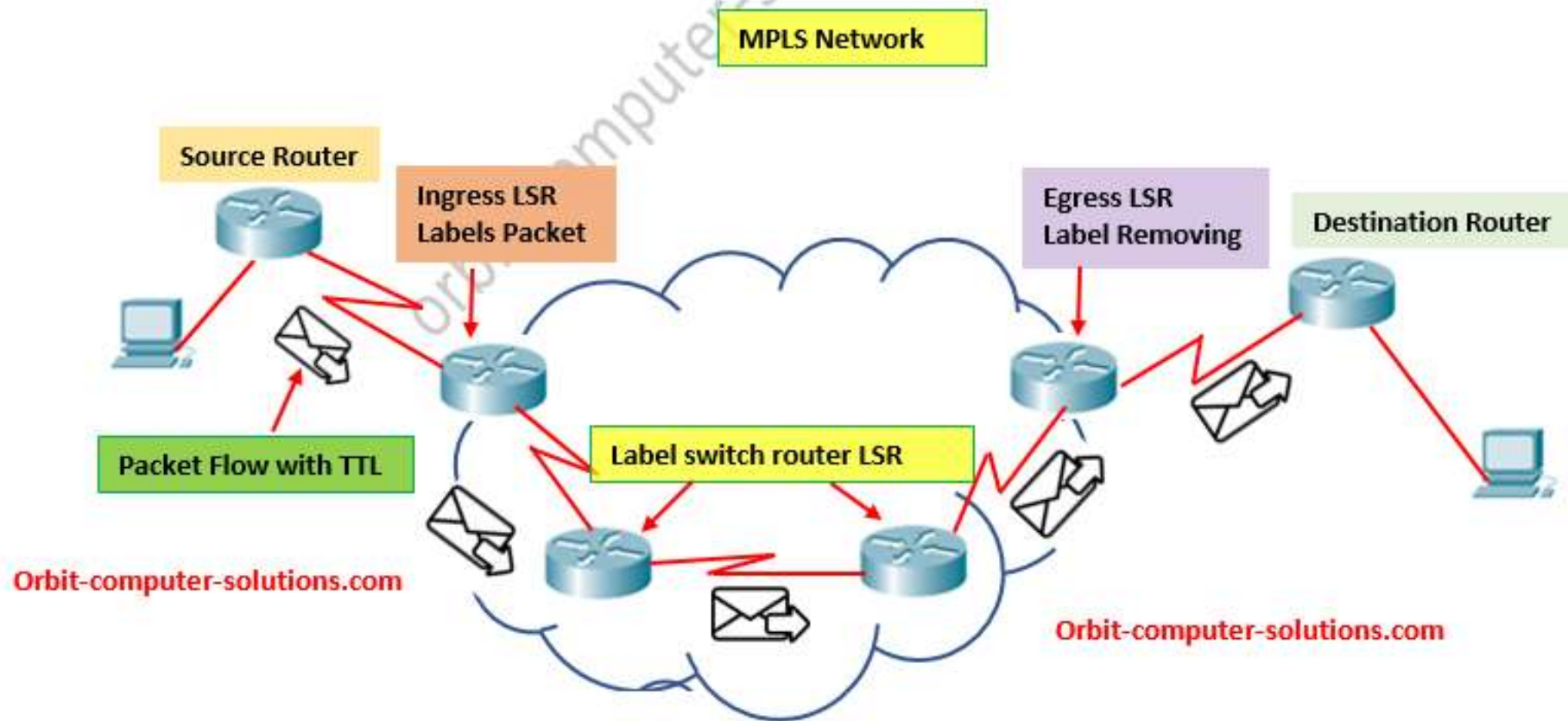
- MPLS operates at a layer that is generally considered to lie between OSI Layer 2 (data link layer) and Layer 3 (network layer), and thus is often referred to as a *layer 2.5 protocol*.
- MPLS allows IP packets to be forwarded at layer 2 switching level without being passed up to layer 3: routing level.
- As we know routing is the layer 3 function and switching is the layer 2 function, MPLS makes those *routers on the internet act like switches on LAN*. That's why MPLS is called 2.5 protocol.
- It was designed to provide a unified data-carrying service for both circuit-based clients and packet-switching clients which provide a datagram service model. It can be used to carry many different kinds of *traffic*, including IP packets, as well as native ATM, SONET, and Ethernet frames.
- <https://www.youtube.com/watch?v=BuIWNecUAE8>

MPLS Architecture



...

<https://www.youtube.com/watch?v=U1w-b9Glt0k>



MPLS benefits

- Elimination of redundancies
- Improve efficiency
- Reduce cost
- Improve scalability
- Improve reliability

xDSL

- xDSL refers to the sum total of digital subscriber line (DSL) technologies.
- Line-length limitations on DSL signal transmissions from the telephone exchanges speeds have resulted in the many types of DSLs.
- xDSL ranges from 6.1Mbps to 155Mbps incoming, and from 600Kbps to 15Mbps outgoing.
- The "x" is a wildcard that can be ADSL (Asymmetric) or SDSL (Symmetric).
- xDSL uses digital encoding to provide more bandwidth over existing twisted-pair telephone lines (POTS).
- Many iterations of xDSL allow the phone to be used for data communication at the same time it's being used to transmit data. This is because phone conversations use frequencies below 4KHz, above which xDSL tends to operate. Several types of xDSL modems come with "splitters" for using voice and data concurrently.

xDSL family

- Some examples of DSL technologies (xDSL) include
 - Digital Subscriber line (DSL)
 - Integrated Services Digital Network (**ISDN**)
 - Asymmetric Digital Subscriber Line (**ADSL**)
 - Symmetric Digital Subscriber Line (**SDSL**)
 - Gigabit Digital Subscriber Line (GDSL)
 - High-Data-Rate Digital Subscriber Line (**HDSL**/HDSL2)
 - Rate-Adaptive Digital Subscriber Line (RADSL)
 - Very-High Speed Digital Subscriber Line (VDSL/VDSL2)
 - Universal High-Bit-Rate Digital Subscriber Line (UHDSL)

xDSL family

DSL Type	Download	Upload	Distance (feet)
ADSL (Asymmetrical)	1.5 - 8 Mbps	16 kbps to 640 kbps	9K to 18K
UDSL(a.k.a. G.lite, DSL Lite)	1.5 Mbps	384 kbps	12K to 18K
RADSL (Rate Adaptive)	Variable to 7 Mbps	Variable to 640 kbps	18K to 25K
VDSL (Very High Bit Rate)	26 Mbps to 52 Mbps	3 Mbps to 6 Mbps	1K to 3K
IDSL (ISDN over DSL)	144 kbps	144 kbps	18K (more w/ repeater)
SDSL (Symmetrical)	144 kbps to 2 Mbps	144 kbps to 2 Mbps	11.5K to 22K
HDSL (High Bit Rate)	1.544 Mbps	1.544 Mbps	12K on 2 pairs
HDSL (High Bit Rate)	2.048 Mbps	2.048 Mbps	12K on 3 pairs
HDSL2	1.544 Mbps	1.544 Mbps	12K on 1 pair
SHDSL (Single-pair HDSL)	192 kbps to 2.312 Mbps	3 Mbps to 6 Mbps	1K to 3K

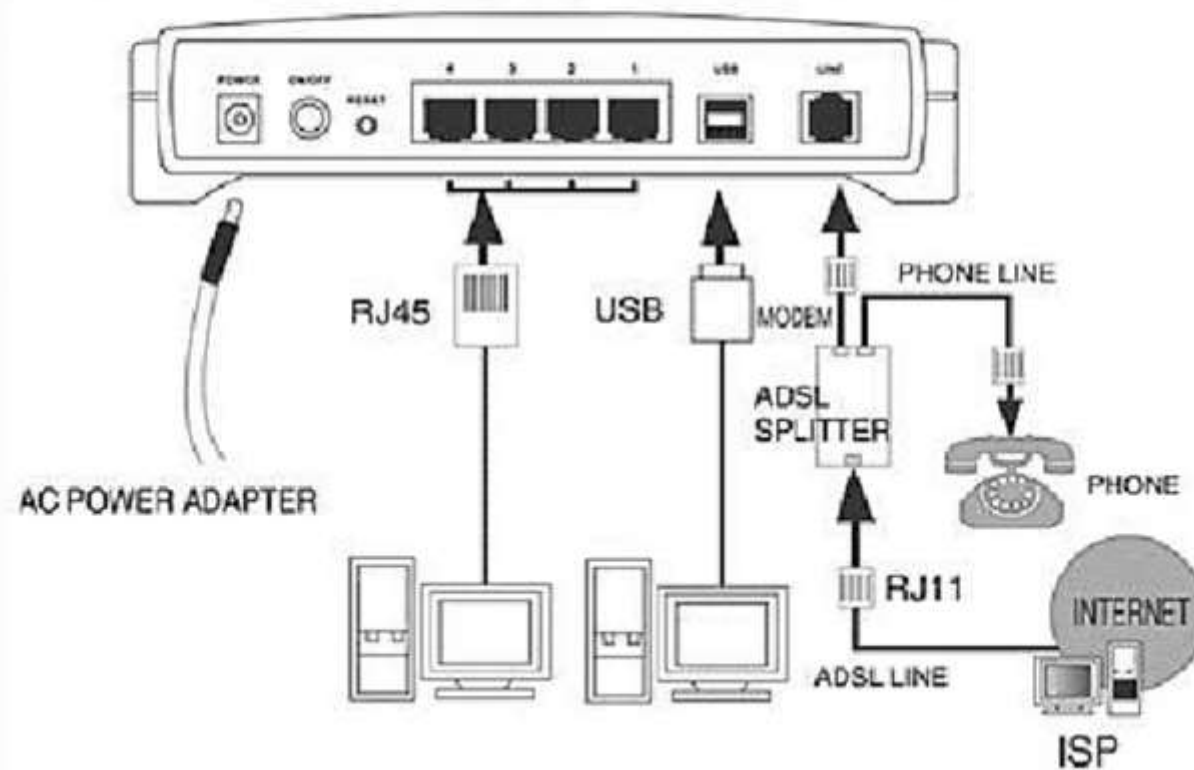
ADSL

- Asymmetric Digital Subscriber Line (ADSL) is asymmetric because of its relatively high capacity to download data when compared to its lower upload capacity
- Uses FDM
- More bandwidth is usually required for downstream transmissions, such as receiving data from a host computer or downloading multimedia files.
- ADSL's asymmetrical nature provides bandwidth where it's needed.
- ADSL provides transmission speeds ranging from downstream/upstream rates of 9 Mbps/640 kbps (business) over a relatively short distance to 1.544 Mbps/16 kbps (a residential customer) as far away as 18,000 feet.

ADSL

- ADSL allows for a splitter box that lets users talk on the telephone at the same time data is being transmitted.
- ADSL uses carrier less amplitude phase modulation (CAP) or discrete multitone (DMT).
- Asymmetric digital subscriber line service provides an Internet connection at a high bandwidth over a phone line to a home or business.
- ADSL provides an “always on” connection and higher speeds than dial-up Internet (last mile) can provide.

ADSL schematic



R-ADSL

- The speed of a Rate Adaptive Digital Subscriber Line (R-ADSL) is dependent on the capacity of the twisted pair it's running over.
- R-ADSL allows for on-the-fly determinations of proper transmit and receive speeds based upon the quality of the connection, length of the loop, and the type of wire being used in the loop.
- It should be used in situations in which the quality of the line connection is variable or affected by weather.
- R-ADSL also allows for a splitter. It transmits data using CAP.
- **RADSL** is a variation of ADSL in which the equipment can sense how high a bit rate the line can handle and adjust the speed downward accordingly.

SDSL

- Symmetric Digital Subscriber Line (SDSL) is symmetrical in that the incoming and outgoing bandwidth is the same.
- SDSL can duplicate T1 or E1 speeds over normal twisted-pair (copper) phone cable for distances up to 11,000 feet (looped, so you must be within about 1 wire mile of your telephone company).
- SDSL uses carrier less amplitude phase modulation (CAP).
- **SDSL** is a symmetrical form of xDSL, but at roughly half the speed of HDSL and using **one wire pair** instead of two.
- SDSL is generally marketed at business customers, while ADSL is marketed at private as well as business customers.

HDSL

- Unlike ADSL, a high-speed digital subscriber line (HDSL) provides **symmetrical** pathways for bandwidth—matching upstream and downstream circuits—at speeds ranging from **128 kbps to 1.544 Mbps** over distances from **5000 to 18,000 feet**.
- Based on its speed/distance capabilities, HDSL best serves WAN applications and has the potential to serve as a better physical layer for **T1 services**.

HDSL

- HDSL uses the same encoding methods employed by ISDN and employs two sets of phone lines.
- It also employs a higher number of bits per baud. The incoming and outgoing speeds of HDSL are identical.
- It was the first digital subscriber line(DSL) technology to use a higher frequency spectrum over copper, twisted pair cables.
- HDSL was developed to transport DS1 services at 1.544 Mbit/s and 2.048 Mbit/s over telephone local loops without a need for repeaters.
- Successor technology to HDSL includes HDSL2 and HDSL4, proprietary SDSL, and G.SHDSL.

VDSL

- Very High Digital Subscriber Line (VDSL) is a high-speed version of xDSL, but only for short distances, perhaps a mile or less.
- For very short distances, speeds as high as 50 Mbps can be attained.
 - A basic law of physics states that the shorter the distance the higher the speed that can be attained.
- VDSL offers speeds of up to 52 Mbit/s downstream and 16 Mbit/s upstream, over a single flat untwisted or twisted pair of copper wires using the frequency band from 25 kHz to 12 MHz.
- These rates mean that VDSL is capable of supporting applications such as high-definition television, as well as telephone services (voice over IP) and general Internet access, over a single connection.
- The first-generation VDSL standard specified both quadrature amplitude modulation(QAM) and discrete multi-tone modulation(DMT). In 2006, ITU-T standardized VDSL in recommendation G.993.2 which specified only DMT modulation for VDSL2.

VDSL

- Second-generation systems (**VDSL2**; ITU-T G.993.2 approved in February 2006) use frequencies of up to 30 MHz to provide data rates exceeding 100 Mbit/s simultaneously in both the upstream and downstream directions.
- The maximum available bit rate is achieved at a range of about 300 metres (980 ft); performance degrades as the local loop attenuation increases.
- VDSL standards

Version	Standard name	Common name	Downstream rate	Upstream rate	Approved on
VDSL	ITU G.993.1	VDSL	55 Mbit/s	3 Mbit/s	2001-11-29
VDSL2	ITU G.993.2	VDSL2	200 Mbit/s	100 Mbit/s	2006-02-17
VDSL2-Vplus	ITU G.993.2 Amendment 1 (11/15)	VDSL2 Annex Q VPlus/35b	300 Mbit/s	100 Mbit/s	2015-11-06